MAgent Theory 2.0

Gauge-Covariant Agents
and Inherited Organization

Robert C. Dennis

Independent Researcher

October 2026

Abstract

An agent is represented by a local probability law together with mechanisms for receiving records, making predictions, and changing that law. Gauge transformations change its presentation. Interactions communicate selected information between agents and may depend on their current belief and model laws. We formulate these objects on associated statistical bundles, give explicit law-dependent links, and distinguish a normalized comparison likelihood from a direct divergence between peer laws. A coupled two-sector variational dynamics follows from a declared potential and the Fisher metric. Exact conditional-information identities describe what communication preserves and what a receiver supplies through its own model. A parent is characterized by the information required to inherit future evolution and boundary responses. Exact quotient criteria and counterexamples identify this requirement. Under specified conditional reconstruction channels, log-density spectra give an exact general-law completion of the original smooth dynamics, including a declared class of feedback. A finite example with a prescribed smooth erasure gate gives persistent groups of distinct children and exact inherited public evolution. It discards private degrees of freedom while retaining every public child state. A separate nonsmooth extension gives two successive stages of finite-time fusion in a specified lognormal family. Endogenous sufficient-interface selection, further aggregation of distinct public states, recursive formation under the original smooth dynamics, and the derivation of base geometry remain open.

Reading the construction

The mathematical objects are introduced in the order in which they are needed: agent laws and records, interactions, dynamics, predictive compression, and inherited organization. The main results include their decisive proofs. The appendices give the internal history construction, the relation between directional links and a base connection, and a notation table. Secondary calculations are collected in a separate supplement.

Definitions specify the model's objects; constitutive choices specify how those objects interact. A theorem states a consequence of explicit hypotheses. An open problem identifies a further construction that is not supplied by those hypotheses. General probability measures are used whenever measurability suffices. Finite alphabets, smooth statistical families, and vector fibers enter only where an example or a differential calculation requires them. The general definitions assume no Gaussian closure; the fusion example states its lognormal restriction explicitly.

Contents

Abstract i

1 Agents, Possibilities, and Records 1
1.1 Statistical sections . . . 1
1.2 Presentation and probability . . . 2
1.3 Internal records . . . 2
1.4 What correlations describe . . . 3

2 Law-Dependent Interactions 5
2.1 Two sectors and their comparison spaces . . . 5
2.2 Sending, receiving, and exponential links . . . 6
2.3 An explicit constitutive family for general laws . . . 7
2.4 Dependence, direction, and inherited responses . . . 8

3 Coupled inference on agent laws 9
3.1 A potential for the two interacting sectors . . . 9
3.2 Source selection and reduction . . . 10
3.3 The derivative includes every interaction role . . . 11
3.4 Fisher dynamics with equal mobilities . . . 11
3.5 What a fixed record model changes . . . 12

4 Prediction and Selective Communication 14
4.1 Model presentations and their evaluators . . . 14
4.2 What the receiver reconstructs . . . 15
4.3 A predictive coding criterion . . . 16

5 Inherited Parents and Recursive Formation 18
5.1 Two objects that a parent must retain . . . 18
5.2 Exact deterministic inheritance . . . 18
5.3 Exact stochastic inheritance . . . 20
5.4 Boundary responses and composition . . . 20
5.5 A finite two-level construction . . . 21
5.6 Averaged laws under smooth peer dynamics . . . 22
5.7 Memory and formation . . . 25

6 Finite Fusion with Exact Statistical Inheritance **26**
6.1 A law family closed under geometric averaging . . . 26
6.2 Messages and state-dependent scalar links . . . 27
6.3 The continuous subgradient dynamics . . . 28
6.4 An invariant class with explicit global solutions . . . 29
6.5 Two formation stages without a supplied partition . . . 30
6.6 The inherited handoff and its scope . . . 31

7 Sufficient States for Smooth KL Evolution **33**
7.1 The force that survives marginalization . . . 33
7.2 Conditional reconstruction within the original potential . . . 34
7.3 An exact law-valued completion . . . 36
7.4 Boundary records and feedback . . . 37
7.5 What is still required for formation . . . 39

8 State-Controlled Erasure and Persistent Groups **40**
8.1 A homogeneous population with internal gate coordinates . . . 40
8.2 Finite entry into an exactly erasing region . . . 41
8.3 A change of preferred groups without synchronization . . . 42
8.4 The exact inherited state . . . 44

9 The Remaining Formation Problem **46**

A A Normalized Internal Record Process **47**
A.1 Prediction followed by assimilation . . . 47

B From Directional Interaction to Base Transport **49**

C Notation **51**

Chapter 1

Agents, Possibilities, and Records

The purpose of the theory is to describe agents whose interactions can produce further agents. The first distinction is between an agent's probability law and the possibilities represented by that law. The second is between comparing two descriptions and specifying an interaction that changes them. A parent construction must preserve enough of both to continue the dynamics.

1.1 Statistical sections

Let $\mathcal{C}$ be a smooth base of contexts and let $P \to \mathcal{C}$ be a principal G-bundle. A context labels where a description applies; it is not identified with physical position or time. Let K and M be standard Borel spaces, with measurable G-actions by bimeasurable bijections. Their elements k and m represent state contents and model presentations. The actions may be different representations of the same group. They act on probability measures by pushforward.

Definition 1.1 (Statistical agent section). Set $\mathsf{Y} = \mathsf{K} \times \mathsf{M}$, with the product action, and give $\mathcal{P}(\mathsf{Y})$ the sigma-algebra generated by evaluation on measurable sets. The associated statistical bundle is

$$\mathcal{B} = P \times_G \mathcal{P}(\mathsf{Y}), \qquad (ug, \zeta) \sim (u, g_{\#}\zeta). \tag{1.1}$$

The statistical component of agent i is a section ζ_i over a domain $\mathcal{U}_i \subseteq \mathcal{C}$. Its full state also includes the internal variables and access mechanisms read by its update rule.

At a fixed context, disintegration gives

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm), \qquad q_i(dk) = \int q_i(dk \mid m) s_i(dm). \tag{1.2}$$

Here q_i is the state marginal and s_i is the law over model presentations. Regular conditional laws exist on the stated spaces, with versions defined up to the appropriate marginal null sets [6]. A model presentation acquires predictive content through a normalized kernel; its name or coordinate value alone supplies no prediction.

The product restriction $\zeta_i = q_i \otimes s_i$ removes dependence between the two sectors within an agent. A product over agents makes a further restriction on represented contents. Neither restriction prevents the update of one law from depending on other laws. Coupling is a property of the dynamics; factorization is a property of a chosen probability description.

Smoothness is an additional hypothesis. When derivatives of laws are required, use a declared regular statistical family stable under the relevant coordinate changes. The whole space $\mathcal{P}(\mathsf{Y})$ is not assigned a smooth manifold structure by notation. Likewise, a collection of pointwise marginals does not determine a probability law on fields over $\mathcal{C}$.

The population at a context consists of the agents whose domains contain that context. Most of the dynamics below is formulated at a common c_*. This retains the ambient bundle and local sections without requiring parallel transport between distinct base points. Independent interactions between agents at c_* are defined in chapter 2. Directional transport within an agent is treated in appendix B.

1.2 Presentation and probability

In local vector coordinates, write $U_i^q : V_q \to E_i^q$ and $U_i^s : V_s \to E_i^s$ for the sector frames. The spaces E_i^q, E_i^s are the represented state and model fibers. A common principal frame can induce both maps through different representations. Reference laws are

$$\bar{q}_i = ((U_i^q)^{-1})_{\#} q_i, \qquad \bar{s}_i = ((U_i^s)^{-1})_{\#} s_i. \tag{1.3}$$

A passive change sends q_i to $(g_i^q)_{\#} q_i$ and U_i^q to $g_i^q U_i^q$, with the corresponding model-sector rule. The reference laws are unchanged. This is a change of description, not an inference step.

For probability measures μ, ν on one measurable space, relative entropy is

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log \frac{d\mu}{d\nu} d\mu, \tag{1.4}$$

when $\mu \ll \nu$, and is $+\infty$ otherwise. In this formula and below, a displayed integral is understood with its declared measure; no common Lebesgue density is required. Relative entropy compares distinguishability of laws [8].

Proposition 1.2 (Coordinate invariance). For a bimeasurable bijection f, including divergences with value $+\infty$,

$$D_{\mathrm{KL}}(f_{\#}\mu\|f_{\#}\nu) = D_{\mathrm{KL}}(\mu\|\nu). \tag{1.5}$$

Proof. Absolute continuity is preserved in both directions. If $h = d\mu/d\nu$, the transformed density is $h \circ f^{-1}$. Integrating its logarithm against $f_{\#}\mu$ gives the original integral. If absolute continuity fails, a set of zero ν-measure and positive μ-measure is carried to a set with the same property. □

Definition 1.3 (Normalized channel). A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ to each x, measurably in x. Its action on a law is

$$(\mu K)(A) = \int K(x, A)\mu(dx). \tag{1.6}$$

A deterministic measurable map $f : X \to Y$ is represented by $K_f(x, \cdot) = \delta_{f(x)}$. Then $\mu K_f = f_{\#}\mu$.

An invertible frame change preserves all distinctions. A selective or noisy channel need not. Both are well-defined operations on general probability laws, but their roles in an agent interaction differ.

1.3 Internal records

At a fixed context, let S_i be the admitted internal-state space of agent i, and let S be the complete configuration space. It contains the statistical laws, source rows, interfaces, and memory variables

that the dynamics reads. Write $\mathrm{pr}_i : \mathsf{S} \to \mathsf{S}_i$ for the local state projection. For the measurable constructions, take these spaces to be standard Borel. A finite collection of probability-law spaces and finite or Euclidean memory spaces supplies such a realization.

Definition 1.4 (Record and readout). A record readout is a measurable map

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i, \tag{1.7}$$

where O_i is the alphabet of internally distinguishable outcomes. If an interaction with source j produces configuration Z_{t+1}, its record variable is

$$O_{ij,t} = \rho_i(\mathrm{pr}_i\, Z_{t+1}). \tag{1.8}$$

For stochastic dynamics this is a measurable function from the probability space of realizations to O_i. A realized value is denoted $o_{ij,t}$. The source label is part of the record only when the internal state retains it.

A record can be transient. Persistence means that the distinction remains accessible through a specified later readout. A list kept outside the agent does not supply this accessibility. A memory coordinate is therefore an explicit part of the model unless the required distinction is already represented in its state.

Proposition 1.5 (Induced record law). Let $\Gamma_{ij} : \mathsf{S} \rightsquigarrow \mathsf{S}$ be a normalized interaction transition. Then

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\left(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\right), \tag{1.9}$$

is a normalized record kernel. If $\Gamma_{ij}(x, \cdot) = \delta_{F_{ij}(x)}$, the record is the deterministic function $\rho_i \circ \mathrm{pr}_i \circ F_{ij}$.

Proof. The right side is the pushforward of a probability measure through the measurable map $\rho_i \circ \mathrm{pr}_i$. Its total mass is one, and measurability in x follows from the defining property of a kernel. The deterministic case follows by evaluating a Dirac measure. □

This definition introduces no observing agent outside the population. It specifies where an outcome is available and how it can enter later dynamics. It does not derive the transition Γ_{ij} or decide whether its stochasticity represents objective chance. Those are further modeling choices.

For example, a vector comparison can be represented by

$$r_{ij} : E_i^q \times E_j^q \times \mathbb{R}^r \to \mathbb{R}^r, \qquad r_{ij}(k_i, k_j, \epsilon) = A_i k_i - B_{ij} k_j + \epsilon. \tag{1.10}$$

A noise law turns this map into a record kernel on represented contents. A joint content law determines its predicted outcomes. A transition on agent states must still specify how an outcome is registered and how the agent changes. A predictive likelihood and an internal update thus have different domains.

1.4 What correlations describe

Suppose two represented bits have the independent uniform law. Register the event that they agree. Their conditional joint law assigns probability $1/2$ to each of $(0, 0)$ and $(1, 1)$. Each marginal

remains uniform, while the conditional expectation of their product changes from $1/4$ to $1/2$. An exact joint posterior has acquired dependence that its two marginals cannot report.

This example identifies a concrete need for a joint law: predicting an observable that reads both bits after the shared record. It does not imply that every admissible agent dynamics must propagate that posterior. A coupled product-family dynamics is another precise model, provided it is declared. Similarly, two routes that inspect the same source must share its identity; replacing it by independent redraws specifies a different experiment.

The configuration x of law-valued agents, a sample tuple $(k_i, m_i)_i$ used in a probabilistic model, and an ensemble of possible configurations are three different objects. A deterministic dynamics on x needs no probability law over configurations. A preparation ensemble can be introduced to assess predictions or compression, but it is then stated explicitly. These distinctions will matter when a parent retains a law over contents while discarding information needed to evolve that law.

Chapter 2

Law-Dependent Interactions

An interaction compares or transmits information between agents. Its law may depend on the agents' current statistical states. Gauge covariance specifies how this dependence is represented when coordinates change; it does not require the interaction to be a frame identification or select a unique mechanism. We first construct these interactions at one context. Transport between different contexts is a further construction, developed in appendix B.

2.1 Two sectors and their comparison spaces

Fix a context c and a finite set of agents defined there. Write

$$\nu_i^q = q_i, \qquad \nu_i^s = s_i, \tag{2.1}$$

for the belief and model marginals of agent i. The sector index a takes values in $\{q, s\}$. These laws live on standard Borel spaces E_i^a. No product assumption between the two sectors, or between different agents, is required to define an interaction between their marginals. Such a product is a separate choice when a joint recognition law is needed.

In a vector realization choose reference spaces V_q, V_s and invertible presentation maps

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{q}_i = (U_i^q)_\#^{-1} q_i, \qquad \bar{s}_i = (U_i^s)_\#^{-1} s_i. \tag{2.2}$$

We also write $\bar{\nu}_i^a = (U_i^a)_\#^{-1} \nu_i^a$. The two presentations can come from different representations of the same principal frame. This does not identify the belief and model interaction maps. Their domains, parameters, and dependence on the laws remain separately specified.

A passive re-expression $y_i' = g_i^a y_i$ sends ν_i^a to $(g_i^a)_\# \nu_i^a$ and U_i^a to $g_i^a U_i^a$. Therefore the reference laws in eq. (2.2) are unchanged. A common reference change $U_i^a \mapsto U_i^a h_a$ instead sends every reference law to $(h_a^{-1})_\# \bar{\nu}_i^a$. The first operation changes an agent's displayed coordinates; the second changes the common coordinates in which interaction parameters are written.

Definition 2.1 (Normalized interaction channel). A sector channel from source j to receiver i is a Markov kernel

$$\mathcal{K}_{ij}^a : E_j^a \rightsquigarrow E_i^a. \tag{2.3}$$

It sends a law to $(\nu_j^a \mathcal{K}_{ij}^a)(A) = \int \mathcal{K}_{ij}^a(y, A) \nu_j^a(dy)$. When the channel depends on the agent configuration, this formula is evaluated at that configuration. Covariance requires

$$\mathcal{K}_{ij}^{a\prime}(g_j^a y, A) = \mathcal{K}_{ij}^a(y, (g_i^a)^{-1} A). \tag{2.4}$$

Composition of sending, transmission, and receiving kernels produces another normalized channel by iterated integration. Invertibility is not required. Alternatively, two channels can send the receiver and source laws into a common message space, where their communicated predictions are compared.

Proposition 2.2 (Covariant law comparison). Under eq. (2.4), the divergence

$$D_{ij}^{a} = D_{\mathrm{KL}}(\nu_i^a \| \nu_j^a \mathcal{K}_{ij}^a), \tag{2.5}$$

is unchanged by passive bimeasurable coordinate changes, including when its value is infinite.

Proof. Changing the integration variable in the kernel gives $\nu_j^{a\prime}\mathcal{K}_{ij}^{a\prime} = (g_i^a)_{\#}(\nu_j^a \mathcal{K}_{ij}^a)$. For any bimeasurable bijection g, absolute continuity is preserved and the transformed Radon–Nikodym derivative is $(d\mu/d\eta) \circ g^{-1}$. Its logarithmic integral is therefore unchanged. Failure of absolute continuity is also preserved, giving infinity on both sides. □

This is the coordinate invariance of relative entropy [8]. It makes the comparison intrinsic without asserting that either law is a posterior under a fixed population model.

2.2 Sending, receiving, and exponential links

Choose a message space H_a and reference interfaces

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a. \tag{2.6}$$

Their direct content map is $M_{ij}^a = U_i^a O_i^a T_j^a (U_j^a)^{-1}$. A deterministic channel can push the entire law through this map even when the image is lower dimensional. A density formula for a nonsingular law cannot then be used on the original ambient space without accounting for that change of support.

Definition 2.3 (Law-dependent reference generator). Let $\mathcal{L}_{ij}^a$ be a measurable function of the reference laws, and of any explicitly declared interface data, with values in $\mathrm{End}(V_a)$. Define

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a \exp(L_{ij}^a)(U_j^a)^{-1}. \tag{2.7}$$

The primitive self-comparison is $L_{ii}^a = 0$. For other edges a receiving–sending factorization is $L_{ij}^a = O_i^a(\bar{\nu}_i^a) T_j^a(\bar{\nu}_j^a)$, provided both maps refer to the same H_a and V_a.

The exponential is taken on the reference endomorphism, not on a map with different domain and codomain. These are $\mathrm{GL}(V_a)$ interaction maps. A smaller group acting on the frames can still re-express them covariantly. Interpreting them as parallel transports of a connection for that smaller group would additionally require generators in its represented Lie algebra.

Proposition 2.4 (Endpoint covariance and reference consistency). The links in eq. (2.7) obey

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.8}$$

They are also independent of a common reference rechoice if

$$\mathcal{L}^{a\prime}_{ij}((h_a^{-1})_{\#}\bar{\nu}^a_i, (h_a^{-1})_{\#}\bar{\nu}^a_j) = h_a^{-1} L^a_{ij} h_a, \tag{2.9}$$

where the prime includes the re-expression of any supplied interface data.

Proof. Passive site changes leave the reference laws and generator unchanged, so multiplication by the transformed endpoint frames proves eq. (2.8). Under a reference rechoice, the exponential series gives $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$. Substitution cancels both occurrences of h_a in eq. (2.7). □

For factored generators, $O^a_i \mapsto h_a^{-1} O^a_i$ and $T^a_j \mapsto T^a_j h_a$ suffice. Message-coordinate changes $T^a_j \mapsto r_a T^a_j$, $O^a_i \mapsto O^a_i r_a^{-1}$ cancel within the product.

Proposition 2.5 (Selective generator and invertible transport). For finite matrices $O : H \to V$ and $T : V \to H$, the map OT has rank at most $\dim H$, whereas e^{OT} is invertible. Moreover,

$$e^{OT} v = v \quad (v \in \ker T), \qquad \operatorname{im}(e^{OT} - I) \subseteq \operatorname{im} O. \tag{2.10}$$

Proof. The rank statement follows from factorization. The exponential series and $e^{OT} e^{-OT} = I$ prove invertibility. Every positive power of OT vanishes on $\ker T$, and every such power has image in $\operatorname{im} O$. Summing the convergent series proves the remaining assertions. □

Thus the exponential link retains an identity action in unread directions. It is a different operation from sending only the selective content OTv. The identity self-link also differs from a coarse internal loop: a parent may retain nonidentity loop information inherited from several child interactions.

2.3 An explicit constitutive family for general laws

The preceding definition does not select the functional $\mathcal{L}$. The following choice shows that non-constant, covariant generators can depend directly on laws without restricting them to a Gaussian family.

Definition 2.6 (Moment-dependent receiving and sending). For each sector let $V_a = \mathbb{R}^{d_a}$ and admit laws with finite second moments and positive-definite covariance. In reference coordinates write their means and covariances as μ^a_i and M^a_i. For a supplied scalar λ^a_{ij} on each nonself directed edge, put

$$L^a_{ij} = \lambda^a_{ij} \mu^a_i (\mu^a_j)^T (M^a_j)^{-1}, \qquad O^a_{ij}(t) = \lambda^a_{ij} \mu^a_i t, \qquad T^a_j(v) = (\mu^a_j)^T (M^a_j)^{-1} v. \tag{2.11}$$

Here $H_a = \mathbb{R}$, and $L^a_{ij} = O^a_{ij} T^a_j$ has rank at most one. Set $L^a_{ii} = 0$ separately. The edge coefficient is constitutive data, not a consequence of covariance.

Proposition 2.7 (Equivariance of the moment family). The family eq. (2.11) satisfies eq. (2.9) under every invertible linear reference change.

Proof. Under $U^a_i \mapsto U^a_i h_a$, the moments become $\mu^{a\prime}_i = h_a^{-1} \mu^a_i$ and $M^{a\prime}_j = h_a^{-1} M^a_j h_a^{-T}$. Consequently

$$(\mu^{a\prime}_j)^T (M^{a\prime}_j)^{-1} = (\mu^a_j)^T (M^a_j)^{-1} h_a. \tag{2.12}$$

Multiplication with the transformed receiving column gives $L^{a\prime}_{ij} = h_a^{-1} L^a_{ij} h_a$. □

No fixed Euclidean metric was inserted into this comparison: the source covariance supplies the inverse bilinear form and transforms with the law. Finite moments and invertible covariance are assumptions of this example alone. Centered laws give zero generators in this family, so it does not encode all distinctions between laws.

For a non-Gaussian example in two dimensions, take independent centered Laplace coordinates of variance one and translate their joint law to means e_1 and e_2 for two agents. Both covariances equal I. With identity frames and unit edge coefficients,

$$L_{12} = E_{12}, \qquad L_{21} = E_{21}, \qquad \Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.13}$$

The loop is nontrivial even though both interfaces have rank one. This is a communication loop at one context, not yet the curvature of a base connection.

2.4 Dependence, direction, and inherited responses

Covariance constrains which law-only generators are possible. If a reference transformation fixes both input laws and all supplied data, eq. (2.9) requires its matrix to commute with the resulting generator. Identical centered laws invariant under every orthogonal transformation therefore cannot select a distinguished off-diagonal direction through a fully orthogonally equivariant law-only rule. Directional interfaces can carry additional information, but that information must be declared.

Opposite attending edges are independent constitutive choices. In particular, Θ_{ji} need not equal Θ_{ij}^{-1}, as eq. (2.13) demonstrates. Geometric reversal of the *same* transport is instead its inverse. Endpoint covariance and conjugation of loop products are the familiar transformation laws of lattice link variables [13]; neither imposes an inverse relation between two independently modeled interactions.

When links depend on the current laws, differentiation must include that dependence. At fixed frames a variation of an objective has the form

$$d\mathcal{E} = d_\nu \mathcal{E} + \sum_{a,i,j} \langle d_{L^a_{ij}} \mathcal{E}, D\mathcal{L}^a_{ij}[d\bar{\nu}^a_i, d\bar{\nu}^a_j] \rangle. \tag{2.14}$$

This statement uses a differentiable statistical family and differentiable constitutive maps. It includes both the sender and receiver occurrences. Freezing the link during an update defines another declared dynamics; it is not generally the derivative of the substituted objective. Nor does substituting live recognition laws into a likelihood make that likelihood part of a fixed generative model during recognition variation.

An inherited parent's state must determine its future boundary response, including the law dependence of its generators. If two child configurations with identical retained data give different future responses, those data do not support autonomous closure.

Chapter 3

Coupled inference on agent laws

An interaction between probability laws can be specified directly, without first interpreting it as an observation likelihood. This chapter chooses such an interaction for both belief and model laws, derives its source rows, and supplies a covariant dissipative dynamics. A normalized model of records defines a second construction. Their relation depends on what is held fixed and on whether the interaction compares laws or sampled contents.

3.1 A potential for the two interacting sectors

Fix a context and a finite set of agents. Agent i has state and model spaces $\mathsf{K}_i, \mathsf{M}_i$, both standard Borel. In this chapter its joint law is restricted explicitly to $\zeta_i = q_i \otimes s_i$. Choose a proper model reference r_i and a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$. The reference joint is

$$R_i(dk, dm) = r_i(dm)G_i(m, dk), \qquad \mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i). \tag{3.1}$$

The evaluator couples the sectors even though their recognition law is a product. Where the terms are finite, disintegration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot))s_i(dm). \tag{3.2}$$

This is a conditional relative-entropy identity: the model marginal contributes the first term and the state conditional contributes the second. In particular, the expectation of a log evaluator in this objective is different from the log of the predictive mixture $s_i G_i$.

Let $\mathbf{q} = (q_i)_i$ and $\mathbf{s} = (s_i)_i$. For each allowed directed edge $j \to i$, choose normalized channels $\mathcal{K}^q_{ij}[\mathbf{q}] : \mathsf{K}_j \rightsquigarrow \mathsf{K}_i$ and $\mathcal{K}^s_{ij}[\mathbf{s}] : \mathsf{M}_j \rightsquigarrow \mathsf{M}_i$. Their dependence on the laws is part of the specified interaction. Define

$$\begin{aligned} \nu^q_{ij} &= q_j \mathcal{K}^q_{ij}[\mathbf{q}], \quad D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij}), \\ \nu^s_{ij} &= s_j \mathcal{K}^s_{ij}[\mathbf{s}], \quad D^s_{ij} = D_{\mathrm{KL}}(s_i \| \nu^s_{ij}). \end{aligned} \tag{3.3}$$

The source sets $\mathcal{N}^q_i, \mathcal{N}^s_i$ are finite and nonempty, with positive normalized reference rows π^q_i, π^s_i. Belief routing uses β_i and model routing uses γ_i on their respective sets.

Definition 3.1 (Coupled law potential). For fixed $T_q, T_s > 0$, set

$$\begin{aligned}\mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_{j \in \mathcal{N}_i^q} \beta_{ij} D_{ij}^q + T_q D_{\mathrm{KL}}(\beta_i \| \pi_i^q) \right] \\ &+ \sum_i \left[\sum_{j \in \mathcal{N}_i^s} \gamma_{ij} D_{ij}^s + T_s D_{\mathrm{KL}}(\gamma_i \| \pi_i^s) \right].\end{aligned} \tag{3.4}$$

The reference laws, evaluators, source priors, and temperatures are held fixed in the variations below. The admissible domain consists of laws for which all displayed divergences are finite.

This potential is a constitutive choice at the level of agent laws. No distribution over these law configurations is required to define it. General probability laws enter eq. (3.4); differentiability will require stronger hypotheses. An identity self-channel has $D_{ii}^q = 0$ throughout its domain, with the same convention in the model sector. Independent opposite attending edges remain independent data.

3.2 Source selection and reduction

Proposition 3.2 (Gibbs rows and the exact reduced potential). Hold the agent laws fixed and suppose every source cost is finite. Define

$$\begin{aligned} Z_i^q &= \sum_j \pi_{ij}^q e^{-D_{ij}^q/T_q}, \quad \beta_{ij}^* = \frac{\pi_{ij}^q e^{-D_{ij}^q/T_q}}{Z_i^q}, \\ Z_i^s &= \sum_j \pi_{ij}^s e^{-D_{ij}^s/T_s}, \quad \gamma_{ij}^* = \frac{\pi_{ij}^s e^{-D_{ij}^s/T_s}}{Z_i^s}. \end{aligned} \tag{3.5}$$

These are the unique minimizing rows. With

$$\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z_i^q - T_s \sum_i \log Z_i^s, \tag{3.6}$$

the exact identity is

$$\mathcal{E} = \mathcal{E}_{\mathrm{red}} + T_q \sum_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*) + T_s \sum_i D_{\mathrm{KL}}(\gamma_i \| \gamma_i^*). \tag{3.7}$$

Proof. For one row, $\log \beta_j^* = \log \pi_j - D_j/T - \log Z$. Substitute this expression into $T D_{\mathrm{KL}}(\beta \| \beta^*)$ to obtain the row objective plus $T \log Z$. Summing proves eq. (3.7). Relative entropy is nonnegative and vanishes exactly when its two probability laws agree, which proves uniqueness, including competitors on the simplex boundary. □

On a differentiable family, direct differentiation of the finite sums in eq. (3.5) gives

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta_{ij}^* dD_{ij}^q + \sum_{ij} \gamma_{ij}^* dD_{ij}^s. \tag{3.8}$$

Indeed, $-T_q d\log Z_i^q = \sum_j \beta_{ij}^* dD_{ij}^q$, and the other sector is identical. There is no derivative of an optimized row to add. Every dependence of its cost on the laws still belongs in dD. If a source prior were varied, this formula would also contain $-T_q \sum_{ij} \beta_{ij}^* d\log \pi_{ij}^q$ and its model counterpart.

3.3 The derivative includes every interaction role

To display the variation, choose densities against fixed references on a regular dominated family. Write a for the density of a receiver law and b for that of its compared law. For mass-preserving variations and justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b)da - \int (a/b)db. \tag{3.9}$$

Here the notation $\int f da$ denotes integration against the signed variation of the law, rather than differentiation of the sample coordinate. The formula follows by differentiating $a\log(a/b)$; the additional integral of da is zero.

For a belief interaction the compared-law variation is

$$d\nu_{ij}^q = (dq_j)\mathcal{K}_{ij}^q[\mathbf{q}] + q_j D\mathcal{K}_{ij}^q[\mathbf{q}](d\mathbf{q}). \tag{3.10}$$

The second term differentiates the prescribed channel itself. Thus an agent contributes when it is a receiver, when it is a source, and whenever its law changes a channel elsewhere. For fixed discrete kernels, the incoming contribution from $j \to i$ to a variation at source value v is

$$-\beta_{ij} \sum_u \frac{q_i(u)}{\nu_{ij}^q(u)} \mathcal{K}_{ij}^q(v,u). \tag{3.11}$$

The model sector has the same expression with q, β replaced by s, γ. If the channels are constructed from law-dependent moments or generators, eq. (3.10) also differentiates those moments and generators. Keeping their numerical values fixed computes a different vector field.

An identity self-channel illustrates why the variable identification matters. Here $a = b$ and $da = db$, so eq. (3.9) is zero throughout the state space. Freezing the source as an old snapshot while moving the receiver replaces this zero functional by a divergence from that snapshot. Its first derivative still vanishes at equality, but its curvature and its force after displacement differ. More generally, directed source selection does not make the gradient of eq. (3.4) one-way: a source appears in the energy of every row that attends to it.

3.4 Fisher dynamics with equal mobilities

Theorem 3.3 (Local flow and exact dissipation on positive tables). Suppose the state and model alphabets are finite, all r_i and evaluator entries $G_i(m,k)$ are strictly positive, and the channel entries are twice continuously differentiable functions of the interior probability tables. Assume every compared-law entry is positive there. Let u^a range over all tables $q_i, s_i, \beta_i, \gamma_i$. Equip their

product with the Fisher metric

$$g_u(v,w) = \sum_a \sum_z \frac{v_z^a w_z^a}{u_z^a}, \qquad \sum_z v_z^a = \sum_z w_z^a = 0. \tag{3.12}$$

For every interior initial condition there is a unique maximal interior solution of

$$\dot{u}_z^a = -u_z^a \left(h_z^a - \sum_w u_w^a h_w^a \right), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}. \tag{3.13}$$

Along this solution,

$$\frac{d\mathcal{E}}{dt} = -\sum_a \mathrm{Var}_{u^a}(h^a) \le 0. \tag{3.14}$$

Proof. The stated positivity and regularity make $\mathcal{E}$ twice continuously differentiable in interior simplex coordinates. The right side of eq. (3.13) is therefore locally Lipschitz, giving a unique local solution and its maximal continuation in the interior. Its entries sum to zero within each table. Centering h^a removes any ambiguity from extending $\mathcal{E}$ off the normalized tables. For every tangent vector v, the metric pairing of $u^a(h^a - \mathbb{E}_{u^a} h^a)$ with v^a equals $\sum_z h_z^a v_z^a$. This identifies the Fisher gradient. Finally, substitute eq. (3.13) into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ to obtain the sum of negative variances. Interior continuation does not assert that the trajectory stays away from the boundary for all time. □

All blocks have unit mobility. A persistent slow model sector would have to follow from the interactions and their state, rather than from an inserted small multiplier in its equation. One may instead optimize the source rows at each state and apply the same construction to $\mathcal{E}_{\mathrm{red}}$ and the matter tables alone; eq. (3.8) gives its complete derivative.

For a regular finite-dimensional family with positive-definite Fisher tensor $g_{ab}(\theta)$, the corresponding equation is $\dot{\theta}^a = -(g^{-1})^{ab}\partial_b \mathcal{E}$ [1]. Whenever the derivatives exist, contraction gives $\dot{\mathcal{E}} = -(\partial\mathcal{E})^T g^{-1}(\partial\mathcal{E})$. Bimeasurable re-expression of samples, carrying all kernels and reference laws with it, preserves relative entropy and Fisher score inner products. On a family preserved by that re-expression, the gradient therefore transforms as a tangent vector. These statements concern fixed coordinate changes; coordinates that themselves evolve require their coordinate-velocity term.

3.5 What a fixed record model changes

Choose a proper prior $P_0(dy, dJ)$ on represented contents and source labels, and a normalized record kernel $\mathscr{R}(do \mid y, J)$. Then $\mathbb{P} = P_0\mathscr{R}$ is normalized by integrating the record first. At a declared record density $\ell_o(y, J)$ with $0 < Z_o = \int \ell_o dP_0 < \infty$, its posterior is $\Pi_o = \ell_o P_0 / Z_o$. Substitution proves

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o, \tag{3.15}$$

when the split terms are defined. The right side supplies the extended definition otherwise. This is the variational free energy of a fixed model. A product family for Q is permitted and its block updates remain coupled through the likelihood factors. Each block derivative contains every factor incident on it.

With independent source priors and one likelihood factor reading each label, a factorized source row has cost $C_{ij} = \mathbb{E}_{Q_Y}[-\log \ell_i(o_i \mid Y, j)]$ and optimizer $\beta_{ij} \propto \pi_{ij} e^{-C_{ij}}$. This expected log likelihood differs from both the peer divergence and the likelihood integrated over a prior. For an isolated source record, exact source inference gives $\mathbb{P}(J = j \mid o) \propto \pi_j \mathbb{E}_{P_Y} \ell_j(o \mid Y)$. Equality of these different rows needs additional hypotheses.

Proposition 3.4 (Entropy and the likelihood Jensen gap). On standard Borel spaces, let $k(w \mid v)$ be a density of a normalized kernel into a receiver space with sigma-finite reference λ. Let the receiver law be $q = a\lambda$ and the source law be p. Put $b(w) = \int k(w \mid v)p(dv)$. Assume $0 < b(w) < \infty$ for q-almost every w, $k > 0$ almost everywhere under $q \otimes p$, and finiteness of $\mathbb{E}_q|\log a|$, $\mathbb{E}_q|\log b|$, and $\mathbb{E}_{q\otimes p}|\log k|$. Define $p^w(dv) = k(w \mid v)p(dv)/b(w)$ on this full-measure set. Then

$$\mathbb{E}_{q\otimes p}[-\log k(W \mid V)] = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK) + \int D_{\mathrm{KL}}(p\|p^w)q(dw). \tag{3.16}$$

Proof. The density ratio gives $D_{\mathrm{KL}}(p\|p^w) = \log b(w) - \int \log k(w \mid v)p(dv)$. Integrating against q and using $-\int \log b dq = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK)$ proves the identity. The last term vanishes exactly when $p^w = p$ for q-almost every w. □

The entropy here is relative to the declared reference. A record likelihood normalized in o need not be normalized in the receiver coordinate w, so proposition 3.4 requires that additional kernel interpretation. Even when it applies, the entropy and conditional divergence can affect the dynamics.

There is also an exact obstruction to universal equality. Freeze a channel K and receiver q. Suppose p_1K, p_2K have positive densities that differ on a set of positive q-measure, with both divergences finite. Strict convexity of $-\log$ gives

$$D_{\mathrm{KL}}(q\|(tp_1 + (1-t)p_2)K) < tD_{\mathrm{KL}}(q\|p_1K) + (1-t)D_{\mathrm{KL}}(q\|p_2K), \qquad 0 < t < 1. \tag{3.17}$$

This follows pointwise on the differing-density set and then by integration. By contrast, the expectation $\mathbb{E}_{q\otimes p}e(W, V)$ of any fixed integrable sample energy is affine in p. It cannot equal the divergence at both endpoints and their strict mixture. The obstruction concerns a fixed sample energy; it does not exclude a direct law potential, a frozen-template episode, or an explicitly different probability model whose variables are law points.

Chapter 4

Prediction and Selective Communication

A model law is useful through the records it predicts. Communication can preserve those predictions while discarding other distinctions, and a receiver can combine a message with information already present in its model. This chapter gives exact identities for both operations. The retained prediction is always named; agreement about one output is not automatically agreement about every future interaction.

4.1 Model presentations and their evaluators

Let $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$ be a normalized local model and let $H_i : \mathsf{K}_i \rightsquigarrow \mathsf{O}_i$ be a normalized record kernel. Their composition

$$K_i^{\mathrm{rec}}(m, A) = \int H_i(k, A) G_i(m, dk), \qquad p_i^{\mathrm{rec}} = s_i K_i^{\mathrm{rec}}, \tag{4.1}$$

gives the model's predictive record law. An evaluator may also take a preparation, accessible history, or chosen interaction as input; these are held fixed when two predictions are compared. More general normalized evaluators need not factor through G_i and H_i.

Suppose a model link $\Theta_{ij}^s : \mathsf{M}_j \to \mathsf{M}_i$ is a bimeasurable bijection. Exact compatibility of the two evaluators, after identifying their record alphabets, means

$$K_i^{\mathrm{rec}}(\Theta_{ij}^s m, A) = K_j^{\mathrm{rec}}(m, A). \tag{4.2}$$

Consequently, $s_i = (\Theta_{ij}^s)_{\#} s_j$ implies equal predictive record laws. The compatibility is additional information: two equal laws on a one-point model space can predict opposite deterministic bits through different evaluators.

Theorem 4.1 (Exact predictive information identity). Let s, t be probability laws on a standard Borel model space and let K be a common normalized record kernel. Assume $D_{\mathrm{KL}}(s\|t) < \infty$. Set $\mu(dm, do) = s(dm)K(m, do)$ and $\nu(dm, do) = t(dm)K(m, do)$, and denote their conditional model laws given o by s^o, t^o. Then

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do). \tag{4.3}$$

Proof. The common kernel cancels in the joint Radon–Nikodym derivative, giving $D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(s\|t)$. Disintegrating the same joint laws by their record marginal factors their density ratio into

the record ratio and the conditional model ratio. The logarithm is their sum. Integrating the first term gives $D_{\mathrm{KL}}(sK\|tK)$, and integrating the second first over m gives the displayed conditional term. Finiteness makes the decomposition unambiguous. □

The second term is the model distinction that the record leaves unresolved. Its vanishing is an exact sufficiency condition. A constant evaluator can make all predictions agree while retaining no model distinction at all. Useful agreement therefore requires a specified predictive task as well as a comparison of laws. For a common history kernel, the same identity holds with the full finite record history in place of o.

If the model link depends on the current laws, theorem 4.1 applies to the resulting fixed comparison at that configuration, provided eq. (4.2) holds there. Preservation of compatibility during an update is a separate condition on the dynamics and the evaluator.

4.2 What the receiver reconstructs

A sending map $T_j : E_j \to \mathsf{H}$ induces the message law $\nu_j = (T_j)_{\#}q_j$. Different laws can have the same message law. For linear maps $T_j : E_j \to \mathsf{H}$ and $O_i : \mathsf{H} \to E_i$,

$$\operatorname{rank}(O_i T_j) \le \operatorname{rank}(T_j). \tag{4.4}$$

Writing a message into a larger space cannot by itself recover distinctions discarded by T_j. A richer reconstruction uses the receiver's conditional model.

For a received value h and a declared likelihood $\ell_i(h \mid k)$, Bayesian assimilation is

$$q_i^+(dk) = \frac{\ell_i(h \mid k)\, q_i^-(dk)}{\int \ell_i(h \mid k')\, q_i^-(dk')}, \tag{4.5}$$

when the denominator is finite and positive. An unmeasured coordinate can change because it is conditionally related to the measured one in q_i^-. The observation selects among possibilities already related by the receiver's model.

Proposition 4.2 (Exact conditional reconstruction). Let $T : E \to \mathsf{H}$ be measurable on standard Borel spaces. Let q, p be laws with $D_{\mathrm{KL}}(q\|p) < \infty$, and set $\nu = T_{\#}q$. Choose a regular conditional law $p(dk \mid h)$ for p given $T = h$. Define

$$\widehat{q}(dk) = \int p(dk \mid h)\, \nu(dh). \tag{4.6}$$

Then $T_{\#}\widehat{q} = \nu$ and

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\, \nu(dh). \tag{4.7}$$

In particular, $\widehat{q} = q$ exactly when the two conditional laws agree for ν-almost every h.

Proof. Absolute continuity gives $\nu \ll T_{\#}p$, so the conditional version of p is defined at every h needed up to a ν-null set. That conditional is supported on $T^{-1}(h)$ almost surely. Hence its mixture has pushforward ν. The laws q and $\widehat{q}$ have the same h-marginal and respective conditional laws

$q(\cdot \mid h)$ and $p(\cdot \mid h)$. Factoring their density ratio by h proves eq. (4.7). Nonnegativity and the equality condition of relative entropy give the last claim. □

This reconstruction preserves the communicated marginal and supplies the discarded conditional from p. It makes precise the sense in which a receiver contributes structure that the message alone does not specify. The proposition assumes a common content presentation; a link or a separately declared correspondence is needed before applying it across different fibers.

4.3 A predictive coding criterion

Lossy source coding makes fidelity part of the problem definition [10]. Decoder side information is a realized variable correlated with the source, as in Wyner–Ziv coding [14]; knowing a prior without such a realization does not supply that variable. A few noiseless real numbers also do not specify a finite-bit channel. A communication rate requires a declared coding, precision, or noise model.

For a predictive criterion, let $P(X, Y, Z)$ be a declared preparation law. The source X may be represented content or a complete fine configuration; the choice is fixed in each application. Let Z be accessible receiver information and let Y be a response to be preserved. An encoder $T(du \mid X)$ gives the conditional independence

$$U - X - (Y, Z). \tag{4.8}$$

The decoder predicts Y from (U, Z). This is the setting of an information bottleneck, with the relevant variable explicitly named [11].

Theorem 4.3 (Predictive distortion and the optimal decoder). On standard Borel spaces, assume the displayed information quantities are finite and eq. (4.8) holds. For any predictive decoder $D(dy \mid u, z)$ with finite expected relative entropy,

$$\begin{aligned} &\mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid X, Z) \| D(Y \mid U, Z)\big) \\ &\quad = I(Y; X \mid U, Z) + \mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid U, Z) \| D(Y \mid U, Z)\big). \end{aligned} \tag{4.9}$$

The optimal decoder is $P(dy \mid u, z)$. Its distortion is

$$\mathcal{D}(T) = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{4.10}$$

Proof. Conditional independence gives $P(Y \mid X, U, Z) = P(Y \mid X, Z)$. In the logarithmic ratio defining the left side of eq. (4.9), insert the intermediate conditional law $P(Y \mid U, Z)$. The first resulting expectation is the definition of $I(Y; X \mid U, Z)$. In the second, averaging over X gives the conditional law $P(Y \mid U, Z)$, yielding the remaining divergence. The second term vanishes at the stated decoder and is nonnegative for every decoder. Finally, expand $I(Y; X, U \mid Z)$ in both orders and use $I(Y; U \mid X, Z) = 0$. □

The functional $I(X; U \mid Z) + \lambda \mathcal{D}(T)$, for $\lambda \geq 0$, is one possible criterion for selecting a representation. Its first term is an information cost, not automatically the length of an individual message. An operational coding theorem requires its own source and block-coding assumptions. Transmitting a sample from $(T_j)_{\#} q_j$ and transmitting that whole probability law are also different protocols.

For parent formation, choose X to include the fine configuration and choose Y to be a future boundary response over a specified horizon and interaction. Zero distortion means that U and Z are sufficient for this target under this preparation. A subsequent interaction may read information that this target ignored. Exact inheritance therefore requires the dynamical condition developed in chapter 5, not only predictive sufficiency for one chosen response.

Chapter 5

Inherited Parents and Recursive Formation

A collection becomes a parent when its retained state supports inherited interaction and evolution. A probability law obtained by compression is a first part of this construction. It does not determine how that law will change, what the parent can communicate, or which information its children must retain. We give exact conditions for these further steps and a finite example in which inheritance works through two levels. The example supplies its reductions; formation of the reductions remains a separate question.

5.1 Two objects that a parent must retain

Definition 5.1 (Sample compression and configuration retention). Let $x \in \mathsf{S}$ be a complete agent configuration and let Q_x be its declared law on a standard Borel space Y of represented contents. A normalized sample channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ gives the retained law

$$Q_x^c(A) = \int C(y, A) Q_x(dy). \tag{5.1}$$

A configuration retention is a measurable map $R : \mathsf{S} \to \mathsf{S}^c$. Its value includes the parent laws and the additional data needed for its declared interactions and update.

The distinction applies to both sectors. A parent carrying q_I and s_I also needs the evaluator that makes s_I predictive. Boundary channels, source rows, source marks, and memory can be further components of $R(x)$. Equal marginal laws do not imply equality of these components. When a sample channel reads random links or source variables, those variables belong to the joint input law of the channel. Substitution of a mean link or a selected source is a different retention.

A law of represented contents and a probability law over complete configurations are different objects. Coupled deterministic evolution of law-valued agents requires no probability distribution over x. If such a distribution is used below, its preparation or transition mechanism is stated. All constructions admit general probability laws. A Gaussian parent is one possible family, subject to its own closure conditions.

5.2 Exact deterministic inheritance

Proposition 5.2 (An update descends precisely when it respects retained states). Let S, S^c be measurable spaces, let $R : \mathsf{S} \to \mathsf{S}^c$ be a measurable surjection with a measurable right inverse ι,

and let $F : \mathsf{S} \to \mathsf{S}$ be measurable. There is a measurable parent update F^c satisfying

$$R \circ F = F^c \circ R, \tag{5.2}$$

if and only if

$$R(x) = R(x') \quad \Longrightarrow \quad R(F(x)) = R(F(x')). \tag{5.3}$$

The parent update is unique. Finite spaces automatically admit the required right inverse.

Proof. Equation (5.2) implies the implication in (5.3). Conversely, define $F^c = R \circ F \circ \iota$. It is measurable. Since $R(\iota(R(x))) = R(x)$, the implication gives $F^c(R(x)) = R(F(x))$. Every parent state has a preimage, so this equation also fixes F^c uniquely. □

Repeated substitution gives $R \circ F^n = (F^c)^n \circ R$ for every nonnegative integer n. This is an exact statement about the chosen update, including its order and retained parameters. Replacing that update by descent of a newly defined parent objective requires another argument.

For a differentiable version, suppose R is continuously differentiable and the fine and parent vector fields B, B^c have unique local solutions. Their flows satisfy $R \circ \Phi_t = \Phi^c_t \circ R$ on their common existence interval if

$$DR_x B(x) = B^c(R(x)). \tag{5.4}$$

Indeed, the chain rule makes $R(\Phi_t(x))$ a parent solution with initial value $R(x)$, and uniqueness identifies it. Conversely, differentiation of the flow identity at zero gives (5.4). For a moving map R_t, the left side becomes $\partial_t R_t(x) + DR_t(x)B_t(x)$. Membership changes therefore need an explicit handoff map; a differential identity on one fixed partition does not specify them.

Definition 5.3 (Loss of dynamical roles under product recognition). Take $Q_x = \mathrm{Ber}(p_1) \otimes \mathrm{Ber}(p_2)$, with $x = (p_1, p_2) \in [0,1]^2$, and the coupled deterministic law

$$\dot{p}_1 = -p_1 + \lambda(p_2 - p_1), \qquad \dot{p}_2 = -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \tag{5.5}$$

The linear system has a unique global solution and its vector field points inward on every face of the square. Retain only the law of $K_1 + K_2$. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for the values $0, 1, 2$. However,

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2), \tag{5.6}$$

equals -1.2 and -1.8 respectively. Their retained laws therefore have different derivatives. No autonomous update of that retained law can reproduce both trajectories.

This obstruction occurs while recognition remains a product. The missing datum is which probability belongs to which dynamical role. Correlated recognition is neither a prerequisite for this counterexample nor its automatic remedy. Coupled mean-field dynamics are admissible; the proposed retention must still pass its own inheritance test.

5.3 Exact stochastic inheritance

Proposition 5.4 (Finite strong lumpability). Let T be a transition matrix on a finite configuration space S and let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective. A single parent transition matrix reproduces the retained process for every initial law if and only if, for every $z' \in \mathsf{S}^c$,

$$\sum_{y:R(y)=z'} T(x,y) = \sum_{y:R(y)=z'} T(x',y) \quad \text{whenever } R(x) = R(x'). \tag{5.7}$$

In that case the common sum defines $T^c(R(x), z')$ uniquely.

Proof. The common sums are nonnegative and sum to one, so they define a transition matrix. Conditional on any retained history ending at z, the hidden configuration lies in $R^{-1}(z)$. Every state in that fiber gives the same next retained distribution $T^c(z,\cdot)$; averaging over its conditional law leaves this distribution unchanged. Induction gives the parent path law from the pushed initial law. Conversely, start the fine process at each point mass δ_x. A common parent kernel must give the same one-step distribution whenever these initial states have the same image. This is (5.7). □

This is the finite strong-lumpability criterion [7]. On standard Borel spaces the same argument applies when $T(x, R^{-1}A)$ factors through R as a measurable probability kernel. The qualification is part of the construction. Closure for every initial law is stronger than closure for one chosen preparation.

An equilibrium conditional average of fine transitions defines a normalized candidate parent kernel. It does not prove (5.7): the hidden conditional law during a transient can differ from the equilibrium conditional. Likewise, agreement of invariant distributions does not establish agreement of histories.

5.4 Boundary responses and composition

A parent must inherit what it does to the rest of the population. Fix a declared boundary protocol a, including the preparation and receiving conditions. Let $K_a : \mathsf{S} \rightsquigarrow \mathsf{O}_a$ be its normalized response kernel and let $D_a : \mathsf{O}_a \rightsquigarrow \mathsf{O}_a^c$ retain the response of interest. An inherited kernel K_a^c satisfies

$$(K_a D_a)(x, A) = K_a^c(R(x), A). \tag{5.8}$$

Here $(KD)(x,A) = \int D(o,A)K(x,do)$. Under the measurable right-inverse hypothesis of proposition 5.2, such a kernel exists exactly when $K_a D_a$ is constant on each fiber of R. Define it using the right inverse to prove sufficiency; substitution proves necessity. When several outputs are used jointly, this condition applies to their joint kernel. Preserving each output marginal alone need not preserve their dependence.

For a child channel $\mathcal{K}_{ij} : \mathsf{Y}_j \rightsquigarrow \mathsf{Y}_i$ and retentions $C_j : \mathsf{Y}_j \rightsquigarrow \mathsf{Z}_J$ and $C_i : \mathsf{Y}_i \rightsquigarrow \mathsf{Z}_I$, the corresponding square is $\mathcal{K}_{ij} C_i = C_j \mathcal{K}_{IJ}^c$, where $\mathcal{K}_{IJ}^c : \mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I$. The rightmost expression means first retain the sender and then apply the parent channel. The retained sender must determine every distinction visible to the retained receiver. In a linear realization this reduces to $C_i M_{ij} = M_{IJ}^c C_j$; the kernel-inclusion criterion and its proof are given in the supplement.

Proposition 5.5 (Inheritance composes). Suppose $R_{10} : \mathsf{S}_0 \to \mathsf{S}_1$ and $R_{21} : \mathsf{S}_1 \to \mathsf{S}_2$ intertwine deterministic updates at successive levels. Then $R_{20} = R_{21} \circ R_{10}$ intertwines the first and third updates. The analogous assertion holds for finite lumped Markov transitions. Let $K_a^\ell : \mathsf{S}_\ell \rightsquigarrow \mathsf{O}_\ell$ be boundary kernels, with response retentions $D_{10} : \mathsf{O}_0 \rightsquigarrow \mathsf{O}_1$ and $D_{21} : \mathsf{O}_1 \rightsquigarrow \mathsf{O}_2$. If

$$K_a^0 D_{10} = R_{10} K_a^1, \qquad K_a^1 D_{21} = R_{21} K_a^2, \tag{5.9}$$

where maps on the right are their deterministic kernels, then $K_a^0 D_{10} D_{21} = R_{10} R_{21} K_a^2$.

Proof. For deterministic updates, $R_{21} R_{10} F^0 = R_{21} F^1 R_{10} = F^2 R_{21} R_{10}$, where juxtaposition in this identity denotes function composition. For Markov transitions and their deterministic retention kernels, $T^0 R_{10} R_{21} = R_{10} T^1 R_{21} = R_{10} R_{21} T^2$. For boundary kernels, associate the successive integrals and substitute (5.9). All integrands for events are nonnegative, so interchanging the integration order is valid. □

When changes of gauge presentation act on the retained spaces, the retentions and boundary kernels must transform compatibly with those actions. Re-expressing a commuting square then gives the same square in the new coordinates. Neither a parent frame chosen for this purpose nor the square itself determines a base connection.

5.5 A finite two-level construction

Let a complete configuration be $x = (a, b, c, r) \in \{0, 1\}^4$. The coordinates are labels of internal configurations, with r an internal binary register and $\rho(x) = r$ its readout. To attach actual agent laws, choose

$$q_a = \mathrm{Ber}\left(\frac{1 + 2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2 + a}{4}\right), \qquad \zeta_a = q_a \otimes s_a. \tag{5.10}$$

Thus both sectors vary, with tied parameters in this supplied family; b, c are additional internal marks. A model presentation $m \in \{0, 1\}$ has evaluator

$$K(m, \{1\}) = \frac{1}{4} + \frac{m}{2}. \tag{5.11}$$

The corresponding predictive law is $\nu_a = s_a K = \mathrm{Ber}(1/2 + a/8)$, which distinguishes the two values of a.

At each step draw independent bits $\xi \sim \mathrm{Ber}(1/4)$, $\eta \sim \mathrm{Ber}(1/3)$, and $\upsilon \sim \mathrm{Ber}(1/5)$, independently of preceding draws, and set

$$a' = a \oplus \xi, \qquad b' = b \oplus a \oplus \eta, \qquad c' = c \oplus b \oplus \upsilon. \tag{5.12}$$

Independently of these bit draws, conditional on the current configuration, give r' law ν_a. This supplies a transition of an internal register; the evaluator alone did not force this choice. The symbol $\oplus$ denotes addition modulo two. The dynamics are coupled: a affects the next b, and b affects the next c.

Retain $R_{10}(a,b,c,r) = (a,b,r)$ and then $R_{21}(a,b,r) = (a,r)$. For $f_p(0) = 1-p$ and $f_p(1) = p$, the first parent kernel is

$$\begin{aligned} T^1((a,b,r),(a',b',r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\}) \\ \cdot f_{1/3}(b' \oplus b \oplus a). \end{aligned} \tag{5.13}$$

Summing the fine transition over c' gives this expression, independently of c. Summing it over b' gives $T^2((a,r),(a',r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\})$, independently of b. Both reductions satisfy proposition 5.4. Their configuration spaces have sizes sixteen, eight, and four, and both parents retain the laws in (5.10), their evaluator, and the register. Its readout descends by the identity on record values. The first parent additionally retains b.

Both parents reproduce the entire finite record history as well as the retained configuration history. Indeed, write the fine path probability as its initial mass times its transition factors. Summing over hidden coordinates gives the parent path law by proposition 5.4; the record at each step is already a coordinate of that path. The next record has probability $1/2 + a/8$ of being one, so the inherited response is nonconstant. No independence of the initial coordinates is needed.

This construction has nontrivial reductions and coupled dynamics, but its retentions were specified in advance. It proves exact two-level inheritance, without proving that the dynamics choose either reduction.

5.6 Averaged laws under smooth peer dynamics

The arithmetic mixture of child laws is a normalized candidate parent law. It need not determine either its own derivative or its boundary responses. The following example establishes both failures within the smooth peer-KL family of chapter 3, with identical agents, a coupled evaluator, and equal Fisher mobilities. It concerns this specified retention; it is not an obstruction to every possible parent state.

Definition 5.6 (A coupled binary model with law-dependent channels). Use counting measure on $k, m \in \{-1, 1\}$ and the strictly positive laws

$$q_x(k) = \frac{1+kx}{2}, \qquad s_y(m) = \frac{1+my}{2}, \qquad x, y \in (-1,1). \tag{5.14}$$

Choose $J = 1/2$, reference $r(m) = 1/2$, and normalized evaluator $G_J(k \mid m) = e^{Jkm}/(2\cosh J)$. For the declared product recognition family, the local variational term is exactly

$$\begin{aligned} E_0(x,y) &= \sum_{k,m} q_x(k)s_y(m) \log \frac{q_x(k)s_y(m)}{r(m)G_J(k \mid m)} \\ &= \psi(x) + \psi(y) - Jxy + \log\cosh J, \\ \psi(x) &= \frac{1+x}{2}\log(1+x) + \frac{1-x}{2}\log(1-x). \end{aligned} \tag{5.15}$$

All expectations are finite. The mixed derivative is $-J$, so changing either sector changes the force on the other.

In either sector, let a sender with parameter x use the normalized kernel

$$K_x(\ell \mid k) = \frac{1+\rho(x)k\ell}{2}, \qquad \rho(x) = \frac{1+x^2}{4}, \qquad q_x K_x = q_{f(x)}, \quad f(x) = \frac{x+x^3}{4}. \tag{5.16}$$

Every entry exceeds $1/4$. The channel transmits the sampled sender bit, and its transmission coefficient depends nontrivially on the sender law. For passive frame covariance, present each alphabet as the atoms $\{\pm u_i\}$ of a one-dimensional fiber, with $u_i \neq 0$, and define $\mathcal{K}_{ij}(u_j k, \{u_i \ell\}) = K_{x_j}(\ell \mid k)$. Under $g_i \in \mathrm{GL}(1,\mathbb{R})$, push the laws forward and replace u_i by $g_i u_i$. The reference parameters remain fixed, and

$$\mathcal{K}'_{ij}(g_j v, g_i A) = \mathcal{K}_{ij}(v, A). \tag{5.17}$$

Independent sector frames give the same construction for s; pushing G_J through its model and belief frames preserves its normalization.

For two identical agents, set

$$\begin{aligned} d(x,z) &= D_{\mathrm{KL}}(q_x \| q_z), \\ P(x_1,x_2) &= d(x_1, f(x_2)) + d(x_2, f(x_1)), \\ E &= \sum_{i=1}^{2} E_0(x_i,y_i) + P(x_1,x_2) + P(y_1,y_2). \end{aligned} \tag{5.18}$$

Each source row contains only the other agent, so its normalized weight is one and its row-relative-entropy term vanishes. This is a specialization of the declared peer potential. The local term is the displayed KL expectation; no further identification of the entire law-dependent potential with a fixed-joint-law ELBO is assumed. The product Fisher metric and its negative gradient flow are

$$g = \sum_i \left(\frac{dx_i^2}{1-x_i^2} + \frac{dy_i^2}{1-y_i^2} \right), \qquad \dot{x}_i = -(1-x_i^2)\partial_{x_i} E, \quad \dot{y}_i = -(1-y_i^2)\partial_{y_i} E. \tag{5.19}$$

Both mobilities are one. The vector field is smooth on $(-1,1)^4$.

Proposition 5.7 (The mixture retention fails the dynamical inheritance test). For the flow in (5.19), the retention

$$R(x_1,y_1,x_2,y_2) = \left(\frac{q_{x_1}+q_{x_2}}{2}, \frac{s_{y_1}+s_{y_2}}{2} \right) = (q_{\bar{x}}, s_{\bar{y}}) \tag{5.20}$$

does not support an autonomous parent vector field on the whole interior. The same failure holds if the averaged joint law $\frac{1}{2}\sum_i q_{x_i} \otimes s_{y_i}$ is also retained.

Proof. For $0 < |a| < 1$ and $0 < |c| < 1$, take $A = (0,c,0,c)$ and $B = (a,c,-a,c)$ in the coordinate order of (5.20). Both retain (q_0, s_c) and the same averaged joint law $q_0 \otimes s_c$. These are equal full binary probability laws, not merely equal estimates of their means. The identities $f(-x) = -f(x)$ and $d(-x,-z) = d(x,z)$ imply

$$P(x_1,x_2) = P(x_2,x_1) = P(-x_1,-x_2), \qquad \partial_1 P(a,-a) = -\partial_2 P(a,-a), \quad \nabla P(0,0) = 0. \tag{5.21}$$

At B the two Fisher prefactors are equal. Consequently the peer forces and the two atanh terms cancel in $\dot{\tilde{x}}$, giving

$$\dot{\tilde{x}}(A) = Jc, \qquad \dot{\tilde{x}}(B) = Jc(1-a^2). \tag{5.22}$$

Both model-sector derivatives can also be written exactly. Define

$$h(c) = 2\operatorname{atanh} c - \operatorname{atanh} f(c) + f'(c)\frac{f(c)-c}{1-f(c)^2}. \tag{5.23}$$

Indeed, direct differentiation gives $\partial_x d(x,z) = \operatorname{atanh} x - \operatorname{atanh} z$ and $\partial_z d(x,z) = (z-x)/(1-z^2)$. Thus

$$\begin{aligned} (\dot{y}_1, \dot{y}_2)(A) &= -(1-c^2)(h(c), h(c)), \\ (\dot{y}_1, \dot{y}_2)(B) &= -(1-c^2)(h(c)-Ja, h(c)+Ja), \\ \dot{\tilde{y}}(A) &= \dot{\tilde{y}}(B) = -(1-c^2)h(c). \end{aligned} \tag{5.24}$$

The retained belief derivatives nevertheless differ by $-Jca^2 \neq 0$. For $a = 1/2, c = 1/4$, they are $1/8$ and $3/32$. Hence $DR_x B(x)$ is not constant on a retention fiber, contradicting the necessary condition (5.4). The identical retained joint laws establish the additional assertion. □

The parameter a may be arbitrarily small. A neighborhood of the synchronized configurations therefore does not make this retention exact. Unlike definition 5.3, the example uses identical dynamical roles.

Proposition 5.8 (The same retention loses a boundary response). Choose a child uniformly, draw its belief bit, and transmit that bit through (5.16) to an external receiver. This declared protocol has a normalized binary response law with mean $b(x_1, x_2) = (f(x_1) + f(x_2))/2$. For any m, a with $|m|+|a| < 1$, compare belief parameters (m, m) and $(m+a, m-a)$, keeping both model parameters equal to c. The retained marginal and averaged joint laws agree, but

$$b(m+a, m-a) - b(m, m) = \frac{3}{4}ma^2. \tag{5.25}$$

For $ma \neq 0$, their full binary response laws differ.

Proof. Expand $f(x) = (x + x^3)/4$ and use $((m+a)^3 + (m-a)^3)/2 = m^3 + 3ma^2$. At $m = a = 1/4$, the two response means are $20/256$ and $17/256$. Thus the response kernel is not constant on a retention fiber, violating (5.8). □

Proposition 5.9 (Interior synchronization cannot first occur at finite time). For (5.19), the diagonal $\Delta = \{x_1 = x_2,\ y_1 = y_2\}$ is invariant. A solution starting outside Δ cannot enter it at a finite time while remaining in the interior.

Proof. Agent exchange preserves the potential and metric, hence the vector field. On its fixed set Δ, the corresponding component derivatives agree, which proves invariance. The smooth field has unique local solutions in both time directions. If an interior solution meets Δ, uniqueness from that meeting point identifies it locally, backward as well as forward, with the solution in Δ. Along the original existence interval, the set of times at which it lies in Δ is therefore open; it is also closed by continuity. Connectedness of that interval makes it the entire interval, contradicting the initial condition. □

This result permits asymptotic synchronization, state-dependent threshold selection, and exact quotients that do not require synchronization. It does not prove convergence. An alternative retained state or a restricted invariant family must be tested against both inheritance conditions; the calculations above do not exclude such constructions.

5.7 Memory and formation

When a proposed retention fails, one exact option is to retain more of the configuration. For a declared stochastic process, another is to retain the conditional law of the hidden configuration given the accessible retained history. That conditional law determines future predictions. In a finite model it is a probability vector with an exact recursive update, derived in the supplement. Its attainable values need not form a finite set. Introducing such a filter presupposes a process and a preparation law; it is not compulsory for deterministic coupled agents.

A handoff initializes the parent from the actual fine state or its declared conditional law, including the source marks and memory read by future interactions. When evidence is shared across levels, its likelihood belongs to the joint model once. Reusing a child's record through a complete same-episode parent posterior requires the appropriate conditional or cavity accounting. A subsequent predictive prior is a different operation.

Open problem 5.10 (Inherited recursive formation). Construct a specified population dynamics that selects nontrivial blocks from an admitted class of initially unpartitioned configurations, with a declared persistence criterion and a handoff determined by the state or accessible history. Each selected block must carry inherited belief and model laws, evaluators, boundary responses, and dynamics. The resulting parents must then select a block containing at least two first-stage parents under those inherited interactions, without resetting their source data or supplying a newly tuned attraction at the next level.

Persistence can mean an invariant block relation in deterministic dynamics or a specified path event in stochastic dynamics. The finite construction settles inheritance for a supplied hierarchy. It does not select that hierarchy or establish persistence. An extension with explicit fusion events is constructed in chapter 6; endogenous, exactly inherited recursive formation under the original smooth peer-KL dynamics remains open.

Chapter 6

Finite Fusion with Exact Statistical Inheritance

We now construct a population whose dynamics selects persistent blocks and then fuses the resulting parents. Pairwise fusion penalties occur in convex clustering [5], and suitable nonsmooth consensus protocols produce finite-time agreement [4]. The construction here also specifies the represented laws, model evaluator, gauge transport, parent metric, and boundary kernels. Its peer potential is a new nonsmooth square-root-KL interaction. It is distinct from the smooth KL potential of the preceding dynamics chapter.

6.1 A law family closed under geometric averaging

Work over one context, with state and model spaces $\mathsf{K} = \mathsf{M} = (0, \infty)$. Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of e^Z when $Z \sim N(\mu, \sigma^2)$. An agent of integer multiplicity $n \geq 1$ carries the product law

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad \zeta_n = q_n \otimes s_n. \tag{6.1}$$

Multiplicity is retained state information. The variances are fixed within each type; only the two log means evolve. Laws are first written in a common reference coordinate. A fixed positive frame $U_i^q = U_i^s =: U_i \in \mathrm{GL}^+(1)$ represents both sectors locally by multiplication. Frames do not evolve in this example.

Choose constants $a > c > 0$ and define the normalized reference and evaluator

$$r_n = \mathrm{LN}\left(0, \frac{a}{n(a^2 - c^2)}\right), \qquad G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{6.2}$$

The two occurrences of a lognormal expression in this equation denote ordinary probability laws and kernels. In logarithmic coordinates the joint reference $r_n(dm)G_n(m, dk)$ has precision

$$nA, \qquad A = \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}. \tag{6.3}$$

Its eigenvalues $n(a - c)$ and $n(a + c)$ are positive. The predictive law is

$$s_n G_n = \mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right). \tag{6.4}$$

Thus the evaluator depends on the represented model value. Physical references and evaluators are obtained by pushing both coordinates through the common local frame.

Proposition 6.1 (Local energy and metric). For every $n \geq 1$ and every pair of real means, let

$$C = a - 1 - \frac{1}{2}\log(a^2 - c^2), \qquad \mathcal{U}(\mu^q, \mu^s) = \frac{a}{2}\big((\mu^q)^2 + (\mu^s)^2\big) - c\mu^q\mu^s. \tag{6.5}$$

Then

$$E_{0,n} = D_{\mathrm{KL}}(q_n \otimes s_n \| r_n G_n) = C + n\mathcal{U}, \qquad g_n = n\big((d\mu^q)^2 + (d\mu^s)^2\big). \tag{6.6}$$

Proof. The logarithm is a measurable bijection, so it preserves KL divergence. The recognition law in log coordinates has covariance I_2/n, while the reference has covariance A^{-1}/n. The Gaussian KL formula gives

$$E_{0,n} = \frac{1}{2}\left[\operatorname{tr} A - 2 - \log\det A + n(\mu^q, \mu^s)A(\mu^q, \mu^s)^{\mathsf{T}}\right], \tag{6.7}$$

which is the stated expression. The score for either mean is $n(Z - \mu)$, whose squared expectation is n. The product law has zero cross score expectation, giving the displayed Fisher metric. □

For independent synchronized children of multiplicities n_j, set $n_I = \sum_{j\in I} n_j$. Their retained reference samples are

$$k_I = \prod_{j\in I} k_j^{n_j/n_I}, \qquad m_I = \prod_{j\in I} m_j^{n_j/n_I}. \tag{6.8}$$

Weighted log averages have variance $\sum_j (n_j/n_I)^2/n_j = 1/n_I$. The same calculation for the joint reference gives $r_{n_I}G_{n_I}$. Conditional log observation noise has variance $\sum_j (n_j/n_I)^2/(an_j) = 1/(an_I)$, so the evaluator itself is inherited. Physical samples are first returned through U_j^{-1} and the retained sample is expressed in a chosen parent frame U_I.

6.2 Messages and state-dependent scalar links

Define a normalized message kernel by

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw), \tag{6.9}$$

interpreting $B_1(k, dw) = \delta_k(dw)$. It restores unit log variance: $q_nB_n = \mathrm{LN}(\mu^q, 1)$ and $s_nB_n = \mathrm{LN}(\mu^s, 1)$. These messages give all multiplicities a common comparison family.

Model prediction also commutes with this interface. Put $h = (c/a)^2 + 1/a$ and define $B_n^h(k, dw) = \mathrm{LN}(\log k, h(1 - 1/n))(dw)$, again using the identity at $n = 1$. Both routes in the kernel identity

$$G_nB_n^h = B_nG_1 \tag{6.10}$$

have conditional log mean $(c/a)\log m$ and variance $h - (c/a)^2/n$. Integrating against s_n gives

$$(s_nB_n)G_1 = (s_nG_n)B_n^h = \mathrm{LN}\left(\frac{c}{a}\mu^s, h\right). \tag{6.11}$$

Thus a parent can emit the original predictive message by either normalized route.

For each sector $\nu \in \{q, s\}$, choose a fixed $0 < \chi_\nu < 1$ and set

$$L_{ij}^\nu = \chi_\nu(\mu_i^\nu - \mu_j^\nu), \qquad \Theta_{ij}^\nu = U_i \exp(L_{ij}^\nu)U_j^{-1}. \tag{6.12}$$

Under a local change of frame $U_i \mapsto h_i U_i$, the physical transport changes to $h_i \Theta^\nu_{ij} h_j^{-1}$. The reference means and L^ν_{ij} remain unchanged. This supplies nonidentity, law-dependent positive scalings when means differ. The chosen scalar links have trivial cycle product; no curvature claim is needed.

Transport of the reference message from j shifts its log mean to $\mu^\nu_j + \chi_\nu(\mu^\nu_i - \mu^\nu_j)$. Consequently the directed comparison cost

$$d^\nu_{ij} = \frac{\sqrt{2D_{\mathrm{KL}}\big(\mathrm{LN}(\mu^\nu_i, 1) \| (\exp L^\nu_{ij})_{\#}\, \mathrm{LN}(\mu^\nu_j, 1)\big)}}{1 - \chi_\nu} = |\mu^\nu_i - \mu^\nu_j| \tag{6.13}$$

is gauge invariant. The equality includes the dependence of the link on both endpoint laws. For unequal means, differentiating its log-mean residual gives

$$\partial_{\mu^\nu_i}\big[\mu^\nu_i - \mu^\nu_j - L^\nu_{ij}\big] = 1 - \chi_\nu. \tag{6.14}$$

This factor cancels the denominator in (6.13). Holding the link fixed while differentiating would give a different dynamics.

6.3 The continuous subgradient dynamics

For multiplicities n_i with total $N = \sum_i n_i$, choose $\lambda > 0$ and the potential

$$\mathcal{F}(\boldsymbol{\mu}) = \sum_i n_i\big[C + \mathcal{U}(\mu^q_i, \mu^s_i)\big] + \lambda \sum_{i<j} n_i n_j \big(|\mu^q_i - \mu^q_j| + |\mu^s_i - \mu^s_j|\big). \tag{6.15}$$

All contacts are fixed and present. If source rows are recorded, take $\beta_{ij} = \gamma_{ij} = n_j/N$, including self entries. Summing child entries gives exactly the same parent row. Self comparison costs vanish, while their row mass records the corresponding contact multiplicity. There is no Gibbs row adaptation. The local term has the stated KL origin; no fixed sample-model ELBO identity is asserted for the entire potential.

With $M = \mathrm{diag}(n_1, n_1, \ldots)$, define the flow on $\mathbb{R}^{2r}$, for r current agents, by

$$M\dot{\boldsymbol{\mu}} \in -\partial\mathcal{F}(\boldsymbol{\mu}). \tag{6.16}$$

Solutions are absolutely continuous in the external time $t \geq 0$ and satisfy this inclusion almost everywhere. There are no boundaries or stochastic driving terms. Both law sectors have the same basic mobility; the factor $1/n_i$ comes from their inherited Fisher metric. In coordinates,

$$\dot{\mu}^q_i = -a\mu^q_i + c\mu^s_i - \lambda \sum_{j \neq i} n_j \sigma^q_{ij}, \tag{6.17}$$

$$\dot{\mu}^s_i = -a\mu^s_i + c\mu^q_i - \lambda \sum_{j \neq i} n_j \sigma^s_{ij}, \tag{6.18}$$

where $\sigma^\nu_{ij} = -\sigma^\nu_{ji}$ belongs to $\mathrm{Sign}(\mu^\nu_i - \mu^\nu_j)$ and $\mathrm{Sign}(0) = [-1, 1]$.

Lemma 6.2 (Uniqueness and persistence). Two solutions of (6.16) with the same masses obey

$$\|\boldsymbol{\mu}(t) - \widetilde{\boldsymbol{\mu}}(t)\|_M \le e^{-(a-c)t}\|\boldsymbol{\mu}(0) - \widetilde{\boldsymbol{\mu}}(0)\|_M. \tag{6.19}$$

On any interval where a solution exists, synchronized unit-mass children remain synchronized in both sectors.

Proof. The quadratic part is $(a-c)$-strongly convex in the M metric and the absolute-value part is convex. For subgradients $\xi, \widetilde{\xi}$ at two configurations, monotonicity gives

$$\langle \xi - \widetilde{\xi}, \boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}} \rangle \ge (a-c)\|\boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}}\|_M^2. \tag{6.20}$$

Substitution from (6.16) and integration proves (6.19). Swapping two equal unit-mass children preserves the initial configuration, potential, and metric. Uniqueness therefore preserves their equality. This argument concerns the full two-sector system, not a restricted scalar equation. □

6.4 An invariant class with explicit global solutions

Theorem 6.3 (Finite sticking on a common-offset family). Let all initial masses be one and suppose

$$\mu_i^q(0) = u_i^0 + v_0, \qquad \mu_i^s(0) = u_i^0 - v_0, \tag{6.21}$$

where the u_i^0 are distinct real numbers and $v_0 \neq 0$. There is a unique global solution of the full inclusion for these initial data. Both sectors have the form

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{6.22}$$

Blocks formed by equal u_i persist and follow the mass-weighted version of the same equation.

Proof. Set $\kappa = a - c > 0$. Equal sector orderings reduce (6.18) to

$$\dot{u}_i \in -\kappa u_i - \lambda \sum_{j \neq i} n_j \operatorname{Sign}(u_i - u_j), \qquad \dot{v} = -(a+c)v. \tag{6.23}$$

Introduce the increasing auxiliary time and rescaled positions

$$\tau(t) = \frac{e^{\kappa t} - 1}{\kappa}, \qquad u_i(t) = e^{-\kappa t} z_i(\tau(t)). \tag{6.24}$$

Since $\dot{\tau} = e^{\kappa t} > 0$, direct differentiation gives

$$z_i' \in -\lambda \sum_{j \neq i} n_j \operatorname{Sign}(z_i - z_j). \tag{6.25}$$

An ordered block has constant velocity $\lambda(M_{\text{right}} - M_{\text{left}})$ between collisions. Adjacent blocks approach with relative speed $\lambda(n_I + n_J) > 0$. At their collision, join their masses, retain the common position, and use the new velocity $\lambda(M_{\text{right}} - M_{\text{left}})$. This equals the mass-weighted average of their velocities just before collision. Choosing zero internal subgradients verifies the original inclusion for every constituent child after collision. Each event reduces the number of blocks, so at most

$N-1$ events occur, and the resulting piecewise-linear curve exists for all $\tau \geq 0$. Reversing (6.24) constructs a global full-system solution. Lemma 6.2 makes it unique. □

The class (6.21) has a common difference between its two sector means. This is an initial-data restriction, not an imposed constraint force. It is a proper submanifold of the full parameter space. The two laws of each agent remain distinct at every finite time because $v(t) \neq 0$. The auxiliary τ is a monotone reparameterization of this solution, not a geometric or operational clock.

6.5 Two formation stages without a supplied partition

Theorem 6.4 (Four agents form two parents and then one). In theorem 6.3, take four agents with the unordered initial set

$$\{u_i^0\} = \{-3d, -2d, 2d, 3d\}, \qquad d > 0. \tag{6.26}$$

Two size-two blocks form at t_1 and remain distinct until $t_2 > t_1$, where they fuse under their inherited equations, with

$$t_1 = \frac{1}{\kappa} \log\left(1 + \frac{\kappa d}{2\lambda}\right), \qquad t_2 = \frac{1}{\kappa} \log\left(1 + \frac{5\kappa d}{4\lambda}\right). \tag{6.27}$$

Proof. In sorted order, the initial z velocities are $(3\lambda, \lambda, -\lambda, -3\lambda)$. The two outer gaps close at $\tau_1 = d/(2\lambda)$, leaving parent positions $-3d/2$ and $3d/2$. Their inherited masses are two and their velocities are 2λ and -2λ. Their gap closes after another $3d/(4\lambda)$, giving $\tau_2 = 5d/(4\lambda)$. Inverting (6.24) gives (6.27). □

Exact reflection symmetry is unnecessary. For four sorted initial positions, let their adjacent gaps be $g_1, g_2, g_3 > 0$, and put $H = \max(g_1, g_3)$. If

$$D = g_2 + \frac{g_1 + g_3}{2} - 2H > 0, \tag{6.28}$$

both outer pairs form by $\tau_* = H/(2\lambda)$. To check that no middle collision intervenes, first note that $D > 0$ implies $g_2 > H$, so a middle collision cannot occur first. Suppose $g_1 \leq g_3$. At the first outer collision the middle gap is $g_2 - g_1$; afterward it decreases at speed 3λ until the other outer pair collides at $g_3/(2\lambda)$. Its value there is $D > 0$. The case $g_3 \leq g_1$ is the reflected calculation. The two parent gap then decreases at speed 4λ, and the final time is

$$\tau_{\text{final}} = \tau_* + \frac{D}{4\lambda} = \frac{2g_2 + g_1 + g_3}{8\lambda}. \tag{6.29}$$

Condition (6.28) is strict and defines an open set relative to the ordered common-offset family. Taking all label permutations gives an admitted class with no assigned labels for blocks. The initial laws are unequal across agents, and the interaction rule contains no partition. Membership is selected by attained equality.

6.6 The inherited handoff and its scope

At a collision, restrict a block I to its common means. Summing its local energies and pulling back its metric gives

$$\sum_{i\in I} n_i(C+\mathcal{U}) = n_I(C+\mathcal{U}) = E_{0,n_I} + (n_I-1)C, \qquad g_I = n_I\big((d\mu_I^q)^2 + (d\mu_I^s)^2\big). \tag{6.30}$$

Every external pair term sums to $n_I n_J|\mu_I^\nu - \mu_J^\nu|$. Zero internal subgradients therefore lift each solution of the inherited equation to a fine solution. Uniqueness identifies that lift with the original trajectory. This also proves persistence for inherited masses by expanding a mass into synchronized unit children. With R retaining block means, masses, and frames, the corresponding parameter flows satisfy the following identity. For a fixed formed partition, keep both component index sets fixed even when further components coincide; a later merger is a separate retention composed with R. Then

$$R \circ \Phi_t = \Phi_t^c \circ R \tag{6.31}$$

on every synchronized stratum reached by the construction, for all future times. Weighted geometric averaging is associative, so later collisions give the same retained laws as direct averaging. The state is continuous at a collision; the velocity changes through the inherited mass sum.

Along the constructed continuous trajectory, ordinary differentiation on each interval between collisions gives

$$\frac{d\mathcal{F}}{dt} = -\sum_i n_i\big((\dot\mu_i^q)^2 + (\dot\mu_i^s)^2\big) \quad \text{almost everywhere.} \tag{6.32}$$

The objective is continuous at collisions when the constants above are retained. Comparing with the stationary configuration having every mean zero in (6.19) proves convergence to that configuration for the admitted initial data. No discrete numerical update has been specified, so this calculation makes no discrete-step assertion.

Sample retention is genuinely noninjective: different positive sample vectors have the same geometric mean. For n synchronized unit children, their log samples conditional on the retained log mean $\bar Z$ have law

$$(Z_1,\dots,Z_n) \mid \bar Z \sim N\left(\bar Z\mathbf{1}, I_n - \frac{\mathbf{1}\mathbf{1}^{\mathsf T}}{n}\right). \tag{6.33}$$

The residual is independent of the mean: their joint Gaussian cross covariance is zero. Exponentiation gives a normalized joint reconstruction kernel. Its one-child marginal is (6.9). Independent residuals for all children would be incorrect because the off-diagonal conditional covariances are $-1/n$. Generative joint samples similarly use the covariance of $(\log m, \log k)$ under $s_1 G_1$; their residual kernel is determined by that covariance, not by the recognition product law. The local KL residual $(n-1)C$ records the corresponding state-independent information loss under the reference comparison.

Parameter retention has a narrower meaning. On an exactly synchronized stratum, child means are redundant copies; retaining their common value does not identify different nonsynchronized configurations. Away from that stratum, a block average need not determine its external sign forces. Equation (6.31) makes no claim about such hidden parameter fibers.

The admitted boundary protocol draws fresh messages through B_n, fresh predictive messages through (6.11), or complete fresh represented samples through the joint reconstruction. To type an accessible stored readout, augment the complete population configuration by $o \in \mathsf{O} = (0, \infty)$ and set $\rho(\boldsymbol{\mu}, \boldsymbol{n}, o) = o$. At declared epochs, the transition Γ advances the deterministic parameter flow and overwrites o using the selected normalized scalar output kernel. Between epochs, o is fixed, and it never enters the mean equation. Retention copies this accessible o exactly; it does not claim access to cached private residual samples. For vector readouts the same definition uses the corresponding finite product space and joint kernel. Equal parameter trajectories and conditional output kernels imply equal joint accessible readout laws for every finite sequence of such queries, by multiplying successive conditional kernels. Reusing persistent latent samples or introducing record feedback requires another state specification.

Open problem 6.5 (Formation under smooth peer KL dynamics). The finite mergers above depend on a constitutive nonsmooth interaction. For a locally Lipschitz vector field with an invariant synchronization manifold, an interior trajectory cannot first reach that manifold in finite time: local backward uniqueness would identify its preceding segment with the manifold trajectory. Establishing recursive formation for the smooth KL dynamics therefore requires a different formation criterion or an additional mechanism, together with its own inherited closure calculation.

Chapter 7

Sufficient States for Smooth KL Evolution

The smooth potential of chapter 3 can admit exact reductions while its agents remain different. Such a reduction requires control of the conditional information that enters the retained force. We first identify that information, then construct a closed state made of general probability laws. The construction keeps the original KL potential and unit Fisher mobilities. Its conditional reconstruction channels are additional constitutive data within that potential; they are not a consequence of an arbitrary law-dependent transport. They provide a sufficient-state architecture, but do not yet explain selection of agent groups.

7.1 The force that survives marginalization

Let Q be a positive law on a standard Borel space, let C be a measurable statistic, and write $\mu = C_{\#}Q$. Suppose a differentiable functional has variational derivative h_Q and Fisher velocity

$$\dot{Q} = -Q(h_Q - \mathbb{E}_Q h_Q). \tag{7.1}$$

All identities below are immediate for finite positive tables. On general spaces we assume a dominated differentiable path, square-integrable scores, and integrability that permits the displayed conditional expectations and differentiations. Statements about flow inheritance are on an interval where the fine and proposed retained equations have unique regular solutions. They do not assert unrestricted existence on every space of probability measures.

Proposition 7.1 (Conditional mean force). The retained velocity is

$$\dot{\mu} = -\mu\left(\mathbb{E}_Q[h_Q \mid C] - \mathbb{E}_Q h_Q\right). \tag{7.2}$$

Consequently, an autonomous velocity for μ exists precisely when the centered conditional mean on the right is determined by μ on the admitted class.

Proof. For every bounded measurable f, integrate $f(C)$ against eq. (7.1) and condition on C. This gives the weak form of eq. (7.2). Two laws with the same marginal have the same marginal velocity if and only if their centered conditional forces agree almost everywhere under that marginal. With the stated uniqueness hypothesis, equality of these velocities gives equality of retained flows. □

For $Q(dz, dv) = \mu(dz)\kappa_z(dv)$ and a fixed reference $P(dz, dv) = \nu(dz)\pi_z(dv)$, the local KL force illustrates the missing information:

$$\mathbb{E}_Q[\log(dQ/dP) \mid z] = \log \frac{d\mu}{d\nu}(z) + H(z), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z \| \pi_z). \tag{7.3}$$

Thus a scalar total conditional loss does not usually determine the retained force. Its dependence on z matters. The Fisher metric makes the same separation: if a score is $f(z, v) = a(z) + b(z, v)$, with $\mathbb{E}_\mu a = 0$ and $\mathbb{E}_{\kappa_z} b = 0$, then

$$g_Q(fQ, fQ) = \int a(z)^2 \mu(dz) + \int \left[\int b(z, v)^2 \kappa_z(dv) \right] \mu(dz). \tag{7.4}$$

The cross term vanishes by conditional centering. This calculation is consistent with the statistical behavior of Fisher information under sufficient statistics [2]. Dynamical sufficiency asks a different question: it preserves the specified evolution and responses, and need not preserve inference about every discarded parameter.

7.2 Conditional reconstruction within the original potential

For each agent take $\mathsf{K}_i = \mathsf{Z}_i \times \mathsf{V}_i$ and $\mathsf{M}_i = \mathsf{W}_i \times \mathsf{U}_i$, all standard Borel. Use positive public densities against fixed reference probabilities, so conditional versions share common full-measure public sets. Disintegrate

$$q_i(dz, dv) = \mu_i(dz)\kappa_{i,z}(dv), \qquad s_i(dw, du) = \sigma_i(dw)\lambda_{i,w}(du). \tag{7.5}$$

Neither conditional need be independent of its retained coordinate. Choose fixed private reference probabilities A_i^q, A_i^s and a fixed normalized public reference

$$R_i^c(dz, dw) = r_i^c(dw)G_i^c(w, dz), \qquad R_i = R_i^c \otimes A_i^q \otimes A_i^s, \tag{7.6}$$

with the product reordered into state and model coordinates. The evaluator G_i^c can couple μ_i and σ_i. Independence of the private reference factors is an assumption of this construction, not a restriction of the general theory.

Let $B_{ij}^q[\boldsymbol{\mu}] : \mathsf{Z}_j \rightsquigarrow \mathsf{Z}_i$ be normalized channels, with the regularity needed for the original variation. Define the full channels by

$$\mathcal{K}_{ij}^q[\mathbf{q}]((z_j, v_j), d(z_i, v_i)) = B_{ij}^q[\boldsymbol{\mu}](z_j, dz_i)\kappa_{i,z_i}(dv_i). \tag{7.7}$$

The model channel uses $B_{ij}^s[\boldsymbol{\sigma}]$ and λ_{i,w_i} in the same way. Source sets may exclude self edges; if self comparisons are included, use the primitive identity convention separately. Public channels may be informative and nonconstant functions of the retained laws. For example, on the public space $\{-1, 1\}$, the channel $B_\mu(z, \{z'\}) = (1 + \rho(\mu)zz')/2$, with $\rho(\mu) = (1 + (\mathbb{E}_\mu Z)^2)/4$, is strictly positive, transmits source information, and varies with the law. Public/private maps and reference factors are declared interface data; carrying them through a bimeasurable change of presentation preserves these constructions.

Proposition 7.2 (Exact cancellation of private peer information). Where the divergences are defined,

$$\begin{aligned} q_j \mathcal{K}^q_{ij} &= (\mu_j B^q_{ij})\kappa_i, \\ D_{\mathrm{KL}}(q_i \| q_j \mathcal{K}^q_{ij}) &= D_{\mathrm{KL}}(\mu_i \| \mu_j B^q_{ij}). \end{aligned} \tag{7.8}$$

This is an identity on the full admitted class, including correlated q_i. Its derivative includes the sender, receiver, and channel derivatives.

Proof. Integration over the source conditional gives the first identity. The compared laws have the same receiver conditional $\kappa_{i,z}$, so their density ratio depends only on z. Integrating its logarithm proves the second identity. Differentiating the resulting functional identity on any regular path proves the derivative statement. Holding κ_i fixed only in the denominator would violate that identity and would define another force. □

Let $\mathcal{E}_c$ denote exactly eq. (3.4) applied to $\mu_i, \sigma_i, R^c_i, B^q_{ij}, B^s_{ij}$, with the same rows, source priors, and temperatures. Define

$$\begin{aligned} \ell^q_{i,z} &= \log(d\kappa_{i,z}/dA^q_i), \quad H^q_i(z) = \int \ell^q_{i,z} d\kappa_{i,z}, \\ \ell^s_{i,w} &= \log(d\lambda_{i,w}/dA^s_i), \quad H^s_i(w) = \int \ell^s_{i,w} d\lambda_{i,w}. \end{aligned} \tag{7.9}$$

In the following formulas, assume these log densities are bounded on the time interval in question. This convenient sufficient condition allows arbitrary dominated law shapes and justifies the exponential moments and derivatives used below. We do not impose a Gaussian or finite-dimensional exponential family.

Theorem 7.3 (Exact marginal and conditional dynamics). For the architecture above the original energy is

$$\mathcal{E} = \mathcal{E}_c + \sum_i \int H^q_i(z)\mu_i(dz) + \sum_i \int H^s_i(w)\sigma_i(dw). \tag{7.10}$$

Write $h^q_i = \delta\mathcal{E}_c/\delta\mu_i$ and $h^s_i = \delta\mathcal{E}_c/\delta\sigma_i$, including every interaction role. The full unit-mobility Fisher dynamics gives

$$\begin{aligned} \dot{\mu}_i &= -\mu_i \left[h^q_i + H^q_i - \mathbb{E}_{\mu_i}(h^q_i + H^q_i) \right], \\ \partial_t \ell^q_{i,z} &= H^q_i(z) - \ell^q_{i,z}. \end{aligned} \tag{7.11}$$

The equations for σ_i, ℓ^s_i are obtained by exchanging sectors. Source rows evolve by their original Fisher equations with the public costs, or may be exactly optimized as in proposition 3.2.

Proof. Apply the relative-entropy chain rule to the local reference in eq. (7.6) and use eq. (7.8). This proves eq. (7.10). Up to a constant independent of (z, v), the full derivative with respect to q_i is $h^q_i(z) + \ell^q_{i,z}(v)$. This can be checked directly from the log-density ratio in the local KL; every peer contribution is a functional of μ alone. Conditional averaging gives the first equation of eq. (7.11). Subtract its logarithmic derivative from the full logarithmic derivative of $q_i = \mu_i\kappa_i$. The public score and global centering cancel, leaving the second equation. The private terms contain no source rows, so their variations do not change the row equations. □

The dependence of H_i^q on z is the only private contribution to the retained belief velocity in this architecture. It cannot generally be discarded. Its evolution will require more than its present value.

7.3 An exact law-valued completion

Definition 7.4 (Conditional log-density spectrum). For a private reference A and conditional density κ_z, define

$$\xi_z = (\ell_z)_{\#}A, \qquad \ell_z = \log(d\kappa_z/dA). \tag{7.12}$$

This probability law records how much reference mass carries each log-density value. It forgets which private point carries that value. It obeys

$$\int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{7.13}$$

Theorem 7.5 (Closed conditional-spectrum evolution). For theorem 7.3, the state consisting of all μ_i, σ_i, conditional spectra ξ_i^q, ξ_i^s, and source rows is autonomous. Each spectrum satisfies, in weak form,

$$\frac{d}{dt}\int f(r)\xi_z(dr) = \int f'(r)(H(z) - r)\xi_z(dr), \tag{7.14}$$

for continuously differentiable test functions on the bounded range in use. For a time interval t starting at zero, its exact solution is

$$\begin{aligned} a &= e^{-t}, & Z_z(a) &= \int e^{ar}\xi_z^0(dr), \\ \xi_z^t &= (r \mapsto ar - \log Z_z(a))_{\#}\xi_z^0. \end{aligned} \tag{7.15}$$

The entropy profile obeys

$$\dot{H}(z) = -\operatorname{Var}_{e^r\xi_z}(r). \tag{7.16}$$

Proof. The reference A is fixed. Differentiate $\int f(\ell_z(v))A(dv)$ and substitute the conditional equation in eq. (7.11); this proves eq. (7.14). Direct differentiation shows that

$$\frac{d\kappa_z(t)}{dA}(v) = \frac{\exp(a\ell_z^0(v))}{Z_z(a)} \tag{7.17}$$

solves the normalized conditional equation. Pushing A through its logarithm gives eq. (7.15). Setting $f(r) = re^r$ in the weak equation gives $\dot{H} = H(H+1) - \int (r^2+r)e^r\xi_z(dr)$, which is eq. (7.16). The probability $e^r\xi_z(dr)$ is normalized by eq. (7.13). All private contributions to eq. (7.11) are now functions of the retained spectra. The remaining scores and row equations depend only on retained public data, proving closure. □

This is a configuration retention. Its spectra can be infinite-dimensional, and there is no claim that they give a fixed finite number of parameters. If represented as a sample statistic, the map $v \mapsto \ell_z(v)$ depends on the current law; its motion is exactly the term differentiated in eq. (7.14). It must not be frozen while the law changes. The autonomous equations are inherited equations.

They have not been identified with a new unrestricted Fisher gradient on probability laws over the spectrum coordinate.

Corollary 7.6 (Different internal conditionals with the original public flow). Suppose that, for each agent and each sector separately, the initial spectrum is independent of the public coordinate. Then this property persists, and the public laws and source rows follow exactly the original coarse KL/Fisher system generated by $\mathcal{E}_c$.

Proof. The same initial spectrum gives the same solution of eq. (7.15) at every public coordinate. Consequently $H_i^q(z)$ is constant in z, and $H_i^s(w)$ is constant in w. Each constant disappears under its own marginal centering in eq. (7.11). □

Equal spectra do not require equal conditionals. With a private reference uniform on two points, the conditionals $(3/4, 1/4)$ and $(1/4, 3/4)$ share

$$\xi = \tfrac{1}{2}\delta_{\log(3/2)} + \tfrac{1}{2}\delta_{\log(1/2)}. \tag{7.18}$$

A public bit may select either conditional, producing correlation between public and private variables. Another law with the same public marginal may use the first conditional at both public values and be a product. Both have the same public trajectory in this architecture, although their internal laws differ. Different agents may have different public laws and different spectra. No equality between agents is imposed.

This class is invariant, but it is not reached for the first time at finite interior time. For any $a > 0$, the spectral update in eq. (7.15) is invertible on its image. Indeed, if ξ^t is given, the initial log value is recovered by the map

$$y \longmapsto \frac{y}{a} - \log \int e^{r/a} \xi^t(dr). \tag{7.19}$$

Substituting eq. (7.15) and using $\int e^r \xi^0(dr) = 1$ proves the assertion. Equal spectra at a finite time therefore had to be equal initially. The same inversion applied to eq. (7.17) proves that distinct conditionals cannot first become equal at finite time. Neither fact forbids an autonomous quotient that keeps the unequal spectra.

7.4 Boundary records and feedback

Public-only response kernels descend immediately: the channels B_{ij}^q, B_{ij}^s and evaluator G_i^c are determined by the retained public laws. A private-sensitive response needs more information. In eq. (7.18), let the probabilities of record 1 at the two private points be $(3/4, 1/4)$. The two conditionals yield record probabilities $5/8$ and $3/8$. After record 1, their private posteriors are $(9/10, 1/10)$ and $(1/2, 1/2)$. Thus an unmarked spectrum can fail both the boundary test and the next-update test.

There is an exact completion for a specified family of records. Fix a measurable mark $T : \mathsf{V} \to \mathsf{D}$, with D standard Borel, through which every allowed private readout factors. Retain

$$\eta_z = (\ell_z, T)_{\#} A. \tag{7.20}$$

The correlation between the log-density value and the mark is retained. For a normalized record kernel with strictly positive density $L(o \mid z, d)$, assume the following integrals are finite and nonzero and each updated law stays in the admitted regular domain. A bounded log likelihood is a sufficient local condition for the conditional log-density bounds. Put

$$\begin{aligned} N_z(o) &= \int e^r L(o \mid z, d)\eta_z(dr, dd), \\ N(o) &= \int N_z(o)\mu(dz). \end{aligned} \tag{7.21}$$

Proposition 7.7 (Exact marked feedback). The record density is $N(o)$. Exact conditioning of this sector gives the closed updates

$$\begin{aligned} \mu^+(dz) &= \frac{N_z(o)}{N(o)}\mu(dz), \\ \eta_z^+ &= \big((r, d) \mapsto (r + \log L(o \mid z, d) - \log N_z(o), d)\big)_{\#}\eta_z. \end{aligned} \tag{7.22}$$

Between records, eqs. (7.14) and (7.15) transport the first coordinate of η_z and leave its mark fixed. Hence the marked state determines any finite sequence of these records, feedback updates, and original smooth-flow intervals.

Proof. Since $d\kappa_z = e^{\ell_z} dA$, integration of the likelihood gives eq. (7.21). Bayes' rule gives the displayed marginal update and $d\kappa_z^+/dA = e^{\ell_z} L(o \mid z, T)/N_z(o)$. Taking the logarithm and pushing A gives the marked update. The fine reference remains fixed. The conditional equation after the update is therefore the same equation with this new conditional law. Inductively, the inherited flow gives the next retained state, the inherited record kernel gives the same next-record law, and the inherited update gives the same posterior retained state. Actions selected from retained histories preserve this induction. Any reference or likelihood parameters changed by records must also be retained and updated by a rule that depends only on this state. □

The update is sectorwise. A product of separate belief and model likelihoods preserves the recognition restriction $q_i \otimes s_i$. A likelihood jointly coupling the two sectors need not preserve that product; exact joint conditioning would require a joint recognition state, or a separately declared projection. Marked feedback can also destroy equality of unmarked spectra across public coordinates. The augmented marked state still closes, but the simpler conclusion of corollary 7.6 then need not hold.

For an explicit retained sample law, use

$$Q^c(dz, dr, dd) = \mu(dz)e^r\eta_z(dr, dd). \tag{7.23}$$

It is normalized and reproduces every declared marked record. It is the pushforward under the current, law-dependent statistic $(z, v) \mapsto (z, \ell_z(v), T(v))$. Different assignments of density values to private points within a mark can give the same retained state. If the permitted records distinguish every private point, the mark may have to retain that point, eliminating this compression. The boundary protocol therefore fixes what can be forgotten. Predictive-state constructions likewise

make sufficiency relative to future observations [9]; the formulas here specify the state and update for this particular KL architecture.

7.5 What is still required for formation

Positive source temperature does not give exact isolation by itself. For optimized rows, every allowed finite-cost edge has positive weight. For a dynamic row, set

$$a_{jk} = \log \frac{\beta_j/\pi_j}{\beta_k/\pi_k}. \tag{7.24}$$

The Fisher row equation from eq. (3.4) gives

$$\dot{a}_{jk} = -(D_j - D_k) - T_q a_{jk}. \tag{7.25}$$

Multiplying by $e^{T_q t}$ and integrating shows that finite cost differences keep every such ratio finite on a regular finite interval. Initially positive weights therefore stay positive there. Uniformly bounded cost differences also bound these ratios for all time and prevent asymptotic disappearance of an allowed weight. A formed collective may nevertheless keep nonzero boundary interactions; isolation is not required for inheritance.

The construction above compresses internal conditional information. It does not combine several distinct public agent laws into one smaller collective state. Retaining their complete labeled tuple and calling it a parent would leave that aggregation problem untouched. Nor does recognizing a persistent source-preference group establish a compatible collective retention. The distinction between a coarse observable and an autonomous macroscopic evolution is also expressed by flow intertwining in Barnett and Seth [3].

Open problem 7.8 (Endogenous aggregation with conditional information retained). Within the original smooth potential, construct an initially unpartitioned population whose evolution selects persistent groups, together with a noninjective collective retention that forgets some of their distinct public degrees of freedom while determining all inherited forces, channels, and declared feedback. The conditional-spectrum state may contribute part of that retention. Its existence for each agent does not establish the collective construction, and no subsequent formation stage follows from it alone.

Chapter 8

State-Controlled Erasure and Persistent Groups

There is a conditional first-stage construction inside the original smooth KL potential. A prescribed channel can lose its dependence on some internal laws when a dynamically evolving gate coordinate enters a flat part of that channel. The remaining agents can be distinct, their preferred groups can change, and the retained dynamics can then become exactly autonomous. The mechanism is an engineered constitutive choice. It supplies neither a general explanation of sufficient interfaces nor a second formation stage. The general laws of the preceding chapter remain the framework; the existence example below uses an explicitly invariant family of positive finite laws.

8.1 A homogeneous population with internal gate coordinates

Write $b_x(k) = (1 + kx)/2$ on $\{-1, 1\}$, set $d(x, z) = D_{\mathrm{KL}}(b_x \| b_z)$, and put $\psi(x) = d(x, 0)$. Four agents carry

$$q_i = b_{x_i} \otimes b_{z_i} \otimes b_{h_i}, \qquad s_i = b_{y_i} \otimes b_{w_i} \otimes b_{\widehat{h}_i}. \tag{8.1}$$

The first coordinate is public, the second private, and the third controls a channel. Their unit Fisher metric has diagonal entries $1/(1 - u^2)$ for each mean u. The product preparation within each sector is an assumption. It is invariant under the full table dynamics specified below, rather than a projection imposed after each step.

Take $J = \log 3$ and $h_* = -1/2$. Use the proper reference $r = b_0 \otimes b_0 \otimes b_{h_*}$ and normalized evaluator

$$G_J(k \mid m) = \frac{e^{Jkm}}{2 \cosh J} \tag{8.2}$$

on the public coordinates, with independent private output b_0 and gate output b_{h_*}. The local term is exactly

$$\begin{aligned} E_{0,i} = {} & \psi(x_i) + \psi(y_i) - J x_i y_i + \log \cosh J \\ & + \psi(z_i) + \psi(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*). \end{aligned} \tag{8.3}$$

All agents have the same reference and evaluator. Every nonself agent is an allowed source, with uniform prior and fixed temperature $T > 0$. We use the exact Gibbs-reduced version of the original potential. The bits are reference coordinates: a change of presentation carries the reference laws, channels, and readout maps with it, so the KL and Fisher covariance statements apply.

Define $\chi(r) = 0$ for $r \le 0$, $\chi(r) = 1$ for $r \ge 1$, and, for $0 < r < 1$,

$$\chi(r) = \frac{e^{-1/r}}{e^{-1/r} + e^{-1/(1-r)}}. \tag{8.4}$$

This increasing function is smooth, all its derivatives match at the endpoints, and $M_\chi = \max |\chi'| < \infty$. Fix $h_0 = 1/4$ and $\delta > 0$, and set

$$H_q = \frac{1}{4}\sum_i h_i, \quad H_s = \frac{1}{4}\sum_i \widehat{h}_i, \quad \alpha(H) = \chi((H - h_0)/\delta). \tag{8.5}$$

The gate is a nonlocal function of internal laws. Its threshold and flat region are constitutive data. There is no external time schedule.

Choose gains $0 < \varepsilon, \varrho < 1$. On a belief edge $j \to i$, let the public output kernel be $(1-\varepsilon)b_{x_i} + \varepsilon\delta_{k_{\mathrm{pub}}}$, and independently draw private output from $b_{\alpha(H_q)\varrho z_j}$ and gate output from b_{h_*}. Its compared law is

$$\nu_{ij}^q = b_{(1-\varepsilon)x_i + \varepsilon x_j} \otimes b_{\alpha(H_q)\varrho z_j} \otimes b_{h_*}. \tag{8.6}$$

This normalized channel is strictly positive on the interior and depends smoothly on the current marginal means. It transmits public samples and uses the source's private law. The model channel exchanges (x, z, h) with $(y, w, \widehat{h})$. Define

$$D_\varepsilon(x, v) = d(x, (1-\varepsilon)x + \varepsilon v). \tag{8.7}$$

The original peer cost becomes

$$D_{ij}^q = D_\varepsilon(x_i, x_j) + d(z_i, \alpha(H_q)\varrho z_j) + d(h_i, h_*). \tag{8.8}$$

No fusion cost has been added.

The definitions extend to arbitrary positive tables on each three-bit space by taking the indicated marginal means. At a product table, the derivative of the receiver entropy is additive in its coordinates. Each compared law is a product, and every channel derivative is a function of a single coordinate because the channel parameters depend on marginal means. The local evaluator also gives additive scores within each sector. The full Fisher score is consequently a sum of coordinate scores on eq. (8.1). Differentiating a product shows that the full table flow preserves this family and induces the stated Bernoulli Fisher metric. This argument includes incoming and channel derivatives.

8.2 Finite entry into an exactly erasing region

Lemma 8.1 (Permanent gate closure under the original flow). For sufficiently small $\delta > 0$ and then sufficiently small $\varrho > 0$ relative to δ, there is an open class within eq. (8.1), with

$$|z_i|, |w_i| < p = 3/4, \qquad h_i, \widehat{h}_i \in (h_0 + \delta/2, h_0 + 3\delta/4), \tag{8.9}$$

for which both gates reach zero in finite time and remain zero.

Proof. The exact envelope derivative gives

$$\dot{h}_i = -(1-h_i^2)\left[2(\operatorname{atanh} h_i - \operatorname{atanh} h_*) + \frac{\alpha'(H_q)}{4}\sum_{a,j}\beta^*_{aj}\partial_\alpha d(z_a, \alpha\varrho z_j)\right]. \tag{8.10}$$

The factor two comes from the local term and the receiver's source-independent gate cost. For $|z_a|, |z_j| \le p$ and $\varrho \le 1/2$,

$$\partial_\alpha d(z_a, \alpha\varrho z_j) = \varrho z_j \frac{\alpha\varrho z_j - z_a}{1-\alpha^2\varrho^2 z_j^2}. \tag{8.11}$$

Its absolute value is bounded by $B\varrho$, where one may take $B = 3p^2/[2(1-p^2/4)]$. The four row sums equal one, so the correction to the gate force is bounded by $B\varrho M_\chi/\delta$.

On $K = [1/8, 1/2]$, define the positive constants

$$\begin{aligned} m &= \min_{h\in K} 2(1-h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \\ M &= \max_{h\in K} 2(1-h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \qquad V = M + m/2. \end{aligned} \tag{8.12}$$

Choose $B\varrho M_\chi/\delta \le m/2$. While its coordinates stay in K, each gate coordinate has $\dot{h}_i \le -m/2$ and $|\dot{h}_i| \le V$. Also choose δ so that the initial interval lies in K and

$$\frac{3V\delta}{2m} < h_0 - 1/8. \tag{8.13}$$

The sector mean reaches h_0 by time $3\delta/(2m)$, before any coordinate can leave K. On $H_q = h_0$, both α and α' vanish. Thereafter

$$\dot{h}_i = -2(1-h_i^2)(\operatorname{atanh} h_i - \operatorname{atanh} h_*). \tag{8.14}$$

All these coordinates start above h_* and decrease toward it. Their mean therefore cannot leave the closed-gate region. The model-sector proof is identical.

It remains to justify the private bounds used in the argument. At $\varrho = 0$ the private field is $\dot{z} = -2(1-z^2)\operatorname{atanh} z$, strictly inward at $z = \pm p$. Private derivatives of eq. (8.8) converge uniformly to this field as $\varrho \to 0$, uniformly over normalized source rows and $0 \le \alpha \le 1$. The incoming terms are of order ϱ; differentiating a private mean does not differentiate α. Compactness therefore preserves the strict inward signs for sufficiently small ϱ. The same holds for w. This closes the interval argument. □

Let τ be the later of the two gate-closure times. The construction uses a smooth but nonanalytic channel: its exactly flat region makes finite entry into exact erasure possible. Decreasing a small nonzero analytic gate would not establish this conclusion.

8.3 A change of preferred groups without synchronization

Theorem 8.2 (Persistent heterogeneous pairs). For sufficiently small $\varepsilon > 0$, admit two public states in $(l, u)^2$ and two in $(-u, -l)^2$, with $l = 0.49$, $u = 0.51$, including every assignment of these

signs to the four labels. These boxes are forward invariant. After τ, each agent's unique preferred source in both sectors is the other agent in its box. There are initial private states for which the preferred pairs are instead crossed between these boxes before closure. The public states within each final pair can remain distinct at every finite time.

Proof. For the local public flow, the inequalities

$$Jl - \operatorname{atanh} l > 0, \qquad \operatorname{atanh} u - Ju > 0 \tag{8.15}$$

give strict inward signs on every face of the two boxes. They follow by differentiating $Jv - \operatorname{atanh} v$ near its zero $v = 1/2$. Set $c = (1-\varepsilon)x + \varepsilon v$. Direct differentiation gives

$$\begin{aligned} \partial_1 D_\varepsilon(x,v) &= \operatorname{atanh} x - \operatorname{atanh} c + (1-\varepsilon)\frac{c-x}{1-c^2}, \\ \partial_2 D_\varepsilon(x,v) &= \varepsilon \frac{c-x}{1-c^2}. \end{aligned} \tag{8.16}$$

Both are uniformly $O(\varepsilon^2)$ on $[-u,u]^2$; the first-order terms cancel in the first line. The full public force has one outgoing row and three incoming sources, each weighted by a number in $[0,1]$. A sufficiently small ε preserves every strict inward margin, independently of the private state and the gate.

After gate closure, the private and gate costs in a row do not depend on its source, so Gibbs ranking uses D_ε alone. Taylor's theorem in the first argument of d gives

$$\frac{1}{2}(x-c)^2 \le d(x,c) \le \frac{(x-c)^2}{2(1-u^2)}. \tag{8.17}$$

A same-box source has separation at most $u - l$, whereas an opposite-box source has separation at least $2l$. Since $(u-l)^2/(1-u^2) < (2l)^2$, the same-box partner is the unique preferred source throughout the invariant region. Cross-pair weights remain positive.

For an explicit initial reorganization, choose private means $\pm\zeta$, with $0 < \zeta < p$, in both sectors, assigning each same-private-sign pair one positive and one negative public member. Initially $\alpha \ge 1/2$. The private advantage of the same-sign partner over an opposite-sign source is

$$d(\zeta, -\alpha\varrho\zeta) - d(\zeta, \alpha\varrho\zeta) = 2\zeta \operatorname{atanh}(\alpha\varrho\zeta) \ge \varrho\zeta^2. \tag{8.18}$$

The public cost range is at most $2u^2\varepsilon^2/(1-u^2)$. Choosing this smaller than $\varrho\zeta^2$ makes the initial preferred pairs the crossed pairs. Strict inequalities persist under small perturbations of the initial private values. Thus the final persistent grouping is not an unchanged initial source assignment.

Choose initial public states within a box separated by a fixed positive distance. Their speeds are uniformly bounded, whereas $\tau \le 3\delta/(2m)$. Taking δ sufficiently small keeps them distinct through τ. Afterward the public field is autonomous, smooth, and permutation equivariant. Its equality diagonal for a pair is invariant. Backward uniqueness precludes first entry into that diagonal at a finite time. This proves finite-time distinctness; it does not give a uniform lower bound on their separation as $t \to \infty$. □

The parameter choices are compatible: first choose a sufficiently small δ, then ϱ sufficiently small relative to δ, and finally ε sufficiently small that ε^2 is small relative to ϱ and the fixed public inward margins. All means stay in compact subsets of $(-1, 1)$, giving global continuation in this admitted family.

8.4 The exact inherited state

For $t \geq \tau$ the reduced potential separates exactly:

$$\begin{aligned}\mathcal{E}_{\mathrm{red}} &= \mathcal{E}_{\mathrm{pub}}(\mathbf{x}, \mathbf{y}) \\ &\quad + 2\sum_i [\psi(z_i) + \psi(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*)].\end{aligned} \tag{8.19}$$

Here $\mathcal{E}_{\mathrm{pub}}$ is the original coupled local KL and Gibbs-reduced peer potential with costs D_ε. The product Fisher metric separates as well. Discarding the private and gate laws therefore gives an exact autonomous public system without fitting another potential or mobility.

This independence is generally absent before gate closure. To see that this is an actual change of the vector field, expand one reduced row with $\alpha > 0$ fixed. If $A_{ij} = (x_j - x_i)^2/[2(1 - x_i^2)]$, its first mixed public-private term is

$$\frac{\alpha\varrho\varepsilon^2}{T} z_i \operatorname{Cov}_{j\neq i}(A_{ij}, z_j). \tag{8.20}$$

This follows by expanding $D_\varepsilon = \varepsilon^2 A + O(\varepsilon^3)$, $d(z_i, \alpha\varrho z_j) = \psi(z_i) - \alpha\varrho z_i z_j + O(\varrho^2)$, and the logarithm of the row normalization. The covariance is under the uniform source prior. It is not identically zero even within the crossed-preference preparation. Fix $v = (-b, -a, a, b)$ with $l < a < b < u$, keep $z = \zeta(1, -1, 1, -1)$ fixed, and vary only public $x_i = rv_i$ near $r = 1$. The summed coefficient is

$$\frac{2\zeta^2 r^2}{9}\left[\frac{a^2 - 3b^2 + 6ab}{1 - r^2b^2} + \frac{b^2 - 3a^2 + 6ab}{1 - r^2a^2}\right]. \tag{8.21}$$

Both numerator factors are positive on the stated intervals, so this expression has positive derivative in $r > 0$. Two fixed admissible values of ζ therefore give different public directional gradients at leading order $\varrho\varepsilon^2$. The differentiated remainder is $O(\varrho\varepsilon^3 + \varrho^2\varepsilon^2)$ on these compact sets. For sufficiently small positive gains, equal public data thus have different public derivatives while the gate is open. This expansion establishes nonidentity of the preclosure force; eq. (8.19) is an exact postclosure identity.

For each selected pair I, retain its size and the empirical measure

$$\eta_I = \frac{1}{|I|}\sum_{i\in I} \delta_{(x_i, y_i)}, \qquad M = \sum_I |I|\eta_I. \tag{8.22}$$

This is a deterministic configuration statistic, not an assumed probability distribution over uncertain population configurations. Let $V[v; M]$ be the public Fisher vector field at the public state $v = (x, y)$, computed from $\mathcal{E}_{\mathrm{pub}}$. It is determined by v and the counting measure M: source sums use $M - \delta_v$, and the derivative includes both receiver and incoming source terms from eq. (8.16).

The normalized Gibbs denominator is computed from the same counting measure. Permutation equivariance makes this definition independent of labels, including multiplicities.

Proposition 8.3 (Exact parent evolution and public responses). The inherited parent equations are

$$\frac{d}{dt}\int f(v)\eta_I(dv) = \int \nabla f(v)\cdot V[v;M]\eta_I(dv) \tag{8.23}$$

for every continuously differentiable test function. On the realizable finite-atom states they determine the retained evolution uniquely. They also determine every declared fresh public response whose source selection depends only on public states, including joint responses with a shared sampled child.

Proof. Differentiate the finite sum in eq. (8.22) using the exact public equations. This gives eq. (8.23). Equal retained parent measures admit enumerations that differ only by within-pair permutations. Uniqueness and permutation equivariance of the public system give identical evolved measures, proving the retained uniqueness assertion. For a fresh response, select the child or children using the declared normalized public kernel, then integrate their public sample laws against the retained empirical measures. Both the selection probabilities and sample laws are determined by the retained states. For shared-source outputs, use one shared draw rather than multiplying separate marginals; the same calculation preserves their joint law. □

The parent can carry belief and model marginals $q_I = \int b_x \eta_I(d(x,y))$ and $s_I = \int b_y \eta_I(d(x,y))$, with normalized public evaluator G_J. These two marginals alone are insufficient: η_I retains the paired public roles and receiver-specific routing information. The inherited equations are eq. (8.23), not a newly postulated Fisher flow for the two mixture marginals.

The physical compression here discards the private and gate laws: the admitted family has twenty-four continuous means, and the retained empirical state keeps eight public means, up to within-pair permutations. It keeps every public child state. Cached private readouts, feedback that changes the erased channels, and protocols that later inspect discarded private coordinates are outside the response class. The reference, gate, initial sign-box preparation, and boundary restrictions are all supplied assumptions. The dynamics select changing source preferences and a persistent grouping within them, but do not discover the erasing architecture. The two final sign boxes remain separated, so this example supplies no recursive second-stage formation and no further compression of the distinct public states.

Chapter 9

The Remaining Formation Problem

The constructions describe agents that compare laws, receive internal records, and inherit descriptions across scales. A normalized predictor makes model agreement operational; an inherited update makes a retained description dynamical. Neither follows from identifying a coherent block at one instant.

The quantifiers in open problem 5.10 distinguish a prescribed partition from a selected one, a transient block from a persistent one, and agreement of one record law from agreement of subsequent responses. Likewise, a smaller model-sector mobility imposes a timescale separation without explaining its origin.

Chapter 6 supplies an exact two-stage example after a change of interaction: square-root KL costs produce a nonsmooth force and finite-time equality. Local models, parent Fisher metrics, contact multiplicities, and fresh-response kernels are inherited. The initial class is restricted, and all agents eventually fuse. This mechanism does not establish formation under the smooth potential of chapter 3.

Chapter 7 gives exact general-law reductions under a specified conditional reconstruction architecture. Conditional log-density spectra retain the hidden information entering the marginal force; record marks complete the state for declared feedback protocols. Equal spectra permit different internal conditionals and correlated laws while public marginals follow the original coarse Fisher flow. That equality is an invariant preparation and is not first reached at finite time.

Chapter 8 instead lets an internal gate enter a prescribed flat channel region, after which private laws can be discarded. Source preferences change into persistent pairs whose public states remain distinct at finite times. This first-stage reduction uses a finite product family and no fusion penalty. It supplies the erasing architecture, retains every public child role, and excludes later private-sensitive feedback. Its two invariant public regions do not merge again.

The unresolved collective problem is aggregation of distinct public states under interfaces selected by the dynamics, with inherited responses and updates that permit another formation step. Fine states with identical proposed parent data must have identical retained derivatives and responses. The arithmetic-mixture counterexample shows why this can fail even for identical child roles. Missing information may concern conditional dependence, interface response, or shared-source memory; it need not fit a predetermined distribution family.

Deriving a base geometry also remains separate. State-dependent interactions at one context do not specify compatible transports between nearby contexts. A curved connection requires that additional structure; a common geometry across agents requires compatibility between their structures.

Appendix A

A Normalized Internal Record Process

This appendix constructs a finite-history law from an internal interaction rule. The rule is supplied; normalization does not select it. Let S and O be standard Borel spaces. A state x includes an accessible record register. Write $\rho : \mathsf{S} \to \mathsf{O}$ for its readout.

Definition A.1 (Event and registration). Choose a normalized kernel $R : \mathsf{S} \rightsquigarrow \mathsf{O}$ and a measurable update $A : \mathsf{S} \times \mathsf{O} \to \mathsf{S}$ satisfying

$$\rho(A(x, o)) = o, \tag{A.1}$$

for $R(x, \cdot)$-almost every o, for every admitted x. Define

$$J(x, dx', do) = R(x, do)\delta_{A(x,o)}(dx'). \tag{A.2}$$

If necessary, the state can be explicitly augmented by a register so that eq. (A.1) holds.

Proposition A.2 (Consistent finite histories). For an initial law μ_0 on S, the iterated kernel product

$$\mathbb{P}_n(dx_0, do_1, dx_1, \ldots, do_n, dx_n) = \mu_0(dx_0) \prod_{t=1}^{n} J(x_{t-1}, dx_t, do_t), \tag{A.3}$$

is a probability law. Marginalizing its last pair gives $\mathbb{P}_{n-1}$, and every record agrees with the readout of the resulting state almost surely.

Proof. The product denotes successive kernel integration, not independent factors. Integrating the last pair gives one because $J(x, \mathsf{S} \times \mathsf{O}) = 1$. Repeating this argument gives total mass one and the stated consistency. Each transition is supported on $x' = A(x, o)$ and hence on $\rho(x') = o$. For a finite history, the intersection of these probability-one events still has probability one. □

Nothing in this construction requires a variable external to the agents. It does require a declared accessible register and an update. A stochastic A is handled by replacing its Dirac kernel with a normalized kernel supported on the registration condition. A deterministic transition is recovered when R is a Dirac kernel as well.

A.1 Prediction followed by assimilation

One possible internal state contains a model law s and a record register. Fix a normalized density $k(o \mid m)$ with respect to a reference measure on O. Let

$$R(s, do) = \left[\int k(o \mid m) s(dm) \right] do, \qquad s^o(dm) = \frac{k(o \mid m) s(dm)}{\int k(o \mid m') s(dm')}. \tag{A.4}$$

The update replaces s by s^o and stores o. The denominator is positive and finite for predictive-almost every record. Defining an arbitrary probability law on the remaining null set completes the measurable rule under the usual conditional-law hypotheses.

Proposition A.3 (The Bayesian history identity). For a fixed latent model m, initial law s_0, and fixed normalized observation density k, sequential prediction and Bayesian assimilation give the record density

$$p(o_1, \dots, o_n) = \int \prod_{t=1}^{n} k(o_t \mid m) s_0(dm). \tag{A.5}$$

Proof. After $t-1$ records, the posterior is proportional to $s_0(dm) \prod_{r<t} k(o_r \mid m)$. The next predictive density is the integral with the product extended to t, divided by the integral up to $t-1$. Multiplying these conditional densities cancels every intermediate normalizer and leaves eq. (A.5). □

This identity applies to the stated fixed observation model. If the likelihood itself is changed using the current state, the product of the actual predictive kernels still defines a normalized process, but it need not equal the fixed-latent expression. Likewise, replacing Bayes' rule by a finite natural-gradient step defines a different assimilation mechanism unless the updates coincide. The normalization of a process and its interpretation as Bayesian conditioning are separate statements.

Appendix B

From Directional Interaction to Base Transport

Links between agents at a single context do not determine transport between neighboring contexts. A connection requires a smooth dependence on the direction and size of a base displacement. This construction can occur within one agent; it does not require collective averaging.

Fix one sector and one agent, suppressing the sector superscript. On a coordinate neighborhood of $\mathcal{C}$, let $U_i(c)$ be a twice continuously differentiable invertible presentation and let $a_i = a_{i,\mu}(c)dc^\mu$ be a continuously differentiable matrix-valued one-form. For a principal G-connection its values must lie in the represented Lie algebra. General endomorphism-valued forms instead describe connections for the extended general linear frame bundle.

Definition B.1 (Infinitesimal directional transport). The proposed transport family has first-order behavior

$$\Theta_i(c+\delta c, c) = U_i(c+\delta c)\exp[-a_{i,\mu}(c)\delta c^\mu]U_i(c)^{-1} + O(|\delta c|^2). \tag{B.1}$$

The first-order generator is linear in the tangent displacement. A single finite matrix at c supplies neither this linear dependence nor the identity value at zero displacement.

Proposition B.2 (Connection and curvature). With the convention $D_i = d + B_i$, eq. (B.1) has connection and curvature

$$B_i = U_i a_i U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i(da_i + a_i \wedge a_i)U_i^{-1}. \tag{B.2}$$

Proof. Expanding the two factors at c gives $\Theta_i = I + (dU_iU_i^{-1} - U_ia_iU_i^{-1})(\delta c) + O(|\delta c|^2)$, which identifies B_i. Acting on a vector-valued function gives the operator identity $d + B_i = U_i(d + a_i)U_i^{-1}$. Squaring it yields $(d + B_i)^2 = U_i(d + a_i)^2U_i^{-1}$, and the squares act by the curvature two-forms in eq. (B.2). □

For an explicit curved example, take $U_i = I$ over an open subset of $\mathbb{R}^2$ and set

$$a_{i,x} = E_{12}, \qquad a_{i,y} = E_{21}. \tag{B.3}$$

Both are rank-one receiving–sending products, while their derivatives vanish. Nevertheless,

$$F_{i,xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0. \tag{B.4}$$

Thus even constant directional interfaces can give curvature through noncommutativity. By contrast, replacing a finite pointwise map by a new frame $V(c)$ and setting $B = -dVV^{-1}$ gives zero curvature: it is the case $a = 0$ of eq. (B.2).

Finite paths use the solution of

$$\frac{d}{dt}T_i(t) = -B_i(\dot{\gamma}(t))T_i(t), \qquad T_i(0) = I. \tag{B.5}$$

Uniqueness of this linear equation gives composition under concatenation and inversion under path reversal. The solution is the path-ordered exponential; the usual link and Wilson-line constructions use this ordering [12]. Independently declared opposite attending edges are not two orientations of this same path and need not be inverses.

Proposition B.3 (Compatibility under re-expression). Under an agent-coordinate change $U_i' = g_i U_i$ with the same reference form,

$$B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}. \tag{B.6}$$

Under a change of reference presentation $U_i' = U_i h$, keeping the same intrinsic transport instead requires

$$a_i' = h^{-1} a_i h + h^{-1} dh. \tag{B.7}$$

Proof. Insert $g_i U_i$ into eq. (B.2) and apply the product rule. Inserting $U_i h$ and eq. (B.7) cancels the two terms containing dh, leaving the original B_i. □

The same affine law must hold between overlapping coordinate patches. A directional interface product transforming only by conjugation is a tensorial one-form. It can describe the difference between two connections, but it is not by itself the full affine connection data. Constructing displacement links with the endpoint transformation law and then differentiating them is one way to obtain the required inhomogeneous term.

If patch coordinates satisfy $v_\alpha = g_{\alpha\beta} v_\beta$, their transition functions must obey $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. The displayed affine law then makes the local connections compatible. A nontrivial bundle need not admit a single global presentation U_i. Even on a trivial bundle, zero curvature does not imply path independence: the scalar connection $B = b\,d\theta$ on the circle is flat but has holonomy $e^{-2\pi b}$, which differs from one for real $b \neq 0$. Global path independence requires trivial holonomy as well as local flatness.

Different agents may carry different connections on the same bundle. They represent one common connection only when their local forms satisfy its overlap relations. Neither case requires an averaging construction.

Open problem B.4 (Deriving directional transport from agent interactions). Specify a constitutive family of displacement links from the agents' laws and interfaces that has identity at zero displacement, a smooth first derivative linear in displacement, and compatible endpoint laws on overlaps. Its derivative then determines the connection by proposition B.2. The unresolved step is the selection and construction of that family; pointwise covariance and the possibility of curvature alone do not determine it.

Appendix C

Notation

Symbol	Meaning
$\mathcal{C}, c_*$	Context base and a selected context.
P, G	Principal bundle and its structure group.
K, M	Represented state contents and model presentations.
$\mathcal{P}(X)$	Probability measures on a measurable space X.
ζ_i, q_i, s_i	Joint agent law and its state and model marginals.
U_i^q, U_i^s	Frame representatives in the two sectors.
$\bar{q}_i, \bar{s}_i$	Laws in common reference coordinates on a trivializing patch.
L_{ij}^q, L_{ij}^s	Law-dependent interaction generators in the vector realization.
$\Theta_{ij}^q, \Theta_{ij}^s$	Invertible source-to-receiver interaction maps.
$\mathcal{K}_{ij}^q, \mathcal{K}_{ij}^s$	Normalized source-to-receiver comparison channels.
T_j^a, O_i^a	Sending and receiving maps in sector $a \in \{q, s\}$.
β_i, γ_i	Belief-sector and model-sector source rows.
G_i, H_i	Local model-to-state kernel and state-to-record kernel.
K_i^{rec}	Normalized model-to-record predictive evaluator.
S, x	Complete agent-configuration space and a configuration.
O$_i$, ρ_i	Record alphabet and internal record readout.
$O_{ij,t}, o_{ij,t}$	Record random variable and one realized value.
Γ_{ij}	Interaction transition on configurations.
$\mathcal{E}$	Declared law-level interaction potential.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model.
F, R, F^c	Fine deterministic update, retention map, and inherited update.
a_i, B_i, F_i	Reference connection form, its local representative, and curvature.
X, U, Z, Y	Source, encoded representation, decoder information, and prediction target in the coding construction.

The symbol F for a discrete update and F_i for curvature are used in separate constructions. A frame U_i^a is an invertible map; the coding variable U is a random representation. A Markov kernel returns a probability measure, whereas an ordinary map returns one value. Kernel composition is written in the order in which a law traverses the kernels.

Bibliography

[1] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998. doi: 10.1162/089976698300017746.

[2] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162:327–364, 2015. doi: 10.1007/s00440-014-0574-8.

[3] Lionel Barnett and Anil K. Seth. Dynamical independence: Discovering emergent macroscopic processes in complex dynamical systems. *Physical Review E*, 108:014304, 2023. doi: 10.1103/PhysRevE.108.014304.

[4] Jorge Cortés. Finite-time convergent gradient flows with applications to network consensus. *Automatica*, 42(11):1993–2000, 2006. doi: 10.1016/j.automatica.2006.06.015.

[5] Toby Dylan Hocking, Armand Joulin, Francis Bach, and Jean-Philippe Vert. Clusterpath: An algorithm for clustering using convex fusion penalties. In *Proceedings of the 28th International Conference on Machine Learning*, pages 745–752, 2011. URL https://icml.cc/2011/papers/419_icmlpaper.pdf.

[6] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

[7] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, New York, 1976. ISBN 9780387901923. Reprint of the 1960 edition, with a new appendix.

[8] Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi: 10.1214/aoms/1177729694.

[9] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001. URL https://arxiv.org/abs/cond-mat/9907176.

[10] Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE International Convention Record*, 7(4):142–163, 1959. URL https://gwern.net/doc/cs/algorithm/information/1959-shannon.pdf.

[11] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method, 1999. URL https://arxiv.org/abs/physics/0004057. Manuscript dated September 30, 1999; arXiv:physics/0004057, posted 2000.

[12] David Tong. Gauge theory. University of Cambridge lecture notes, 2018. URL https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf.

[13] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445.

[14] Aaron D. Wyner and Jacob Ziv. The rate-distortion function for source coding with side information at the decoder. *IEEE Transactions on Information Theory*, 22(1):1–10, 1976. doi: 10.1109/TIT.1976.1055508.